



Deep learning-based 3D reconstruction: a survey

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Abstract

Image-based 3D reconstruction is a long-established, ill-posed problem defined within the scope of computer vision and graphics. The purpose of image-based 3D reconstruction is to retrieve the 3D structure and geometry of a target object or scene from a set of input images. This task has a wide range of applications in various fields, such as robotics, virtual reality, and medical imaging. In recent years, learning-based methods for 3D reconstruction have attracted many researchers worldwide. These novel methods can implicitly estimate the 3D shape of an object or a scene in an end-to-end manner, eliminating the need for developing multiple stages such as key-point detection and matching. Furthermore, these novel methods can reconstruct the shapes of objects from a single input image. Due to rapid advancements in this field, as well as the multitude of opportunities to improve the performance of 3D reconstruction methods, a thorough review of algorithms in this area seems necessary. As a result, this research provides a complete overview of recent developments in the field of image-based 3D reconstruction. The studied methods are examined from several viewpoints, such as input types, model structures, output representations, and training strategies. A detailed comparison is also provided for the reader. Finally, unresolved challenges, underlying issues, and possible future work are discussed.

Keywords 3D Object reconstruction · 3D Shape representation · Deep learning · Computer vision

1 Introduction

In recent decades, image-based 3D reconstruction has played an essential role in many applications, eliminating the need for expensive equipment such as laser scanners. We, as humans, are capable of inferring the exact shape and structure of a previously unseen object within seconds. The idea that machines can do the same has been pursued in the modern era, for which tremendous research has been published and various advancements have been made.

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Given a single or multiple 2D images as input, the task of 3D reconstruction aims to reconstruct the 3D structure and geometry of the object(s) present in a scene of interest. In mathematical terms, given $I = \{I_m, m = 1, \dots, n\}$ as a set of 2D input images, 3D reconstruction aims to produce 3D shape(s) of object(s) present in the scene. The reconstructed shape $\hat{X}$ should be as close as possible to the original shape X .

Learning-based methods, due to their superior performance, have superseded traditional methods in many fields of study, in recent years. Often, these novel techniques are not only more efficient, but also offer new capabilities. The same is true for 3D computer vision and, more specifically, for 3D reconstruction. For example, the proposed deep learning models can be trained end-to-end, eliminating the need to design multiple handcrafted stages. Multitasking is also possible with learning-based methods. As such, a single model can simultaneously predict both the 3D shape and the semantic segmentation of a given picture of a scene, as in Murez et al. (2020). Multitasking has the potential to broaden the use cases of a model while improving feature representation capabilities and accelerating the learning process (Crawshaw 2020). Following the demonstration of the capability of learning-based methods and the advancement of neural network development frameworks, successive research have been published in various fields, including 3D reconstruction. Meanwhile, in a short time, the results of the newly published papers have significantly outperformed previous work. Given this active and rapidly improving area of research, every researcher must conduct a thorough review of these works to gain a comprehensive understanding of the most recent advancements.

Fu et al. (2021), Fahim et al. (2021), and Salvi et al. (2020) are among the recent reviews for deep learning-based 3D reconstruction. Fu et al. (2021), Fahim et al. (2021) only review single image 3D object reconstruction methods. Fu et al. (2021) discusses present challenges, reviews network structures of the proposed methods with their training details, and introduces common evaluation metrics and related datasets. A major strength of this work is that a series of experiments are also implemented to analyze the advantages and disadvantages of the reviewed methods. Fahim et al. (2021) provides many details about each work and contains insightful illustrations and tables. Despite the pros mentioned for the above two works, they only focus on single-image 3D reconstruction, which is unsuitable for a reader who wants to gain an overall perspective on both single- and multiple-image 3D reconstruction. Gao et al. (2019) reviews both single and multiple view approaches. However, the study is limited, and the papers covered are from before 2019. Han et al. (2019) is another work that is detailed and well organized, covering both single and multiple-image approaches. The present challenges and future research directions are well explained. However, it does not include more recent research published after 2019.

This study aims to provide the reader with a comprehensive, structured insight into the latest deep learning-based 3D reconstruction methods. We have tried to review the latest research with minimal ambiguity, covering all the necessary details. Both single-image and multiple-image methods of 3D reconstruction are included. We also discuss various activities that go beyond 3D reconstruction by utilizing it as a downstream task to achieve other objectives. We have covered a decent number of studies that have been published since 2014 in top-ranked journals and conferences.

The sections that proceed are grouped as follows: Sect. 2 discusses basic deep learning-based depth estimation ideas and methods. Sections 3 to 5 review 3D reconstruction methods based on volumetric, point-cloud, and mesh representations, respectively. Section 6 is dedicated to novel implicit representations for 3D reconstruction and covers remarkable research on this topic. While the reviewed works in previous sections focused on single object reconstruction, Sect. 7 reviews works that are capable of multiple object

3D reconstruction and even reconstruction of scenes. Section 8 focuses on learning-based multi-view stereo methods, which are an imperative part of the image-based 3D reconstruction. Section 9 explains how Generative Adversarial Networks (GANs) can be leveraged for 3D reconstruction. Section 10 includes works that use 3D reconstruction as a downstream task to achieve other objectives. The most popular loss functions and public datasets are explained and listed in Sects. 11 and 12. A detailed comparison and discussion of the reviewed algorithms are presented in Sect. 13. In the final section, we provide an overview of the reviewed works and future research directions. Figure 1 provides a visual overview of this research.

2 The primary task: depth estimation

One of the most fundamental problems in 3D reconstruction is estimating the depth of image pixels. The task of depth estimation is referred to as inferring the spatial structure of a scene from 2D images. In fact, the task's goal is to recover the critical spatial information that is lost in the process of 3D to 2D projection when capturing the images. Triangulation techniques and epipolar geometry are used to estimate the depth of pixels in traditional 3D reconstruction methods. However, these methods require knowing both the intrinsic and extrinsic parameters of the camera, and the images need to be captured with small changes in the camera's rotation and translation. Recently, deep learning-based algorithms have been utilized to estimate depth maps from a set of images (Huang et al. 2018) or even a single image (Eigen et al. 2014; Godard et al. 2017). However, the latter is not possible with classic algorithms.

In Eigen et al. (2014), using a double-stream convolutional neural network, a coarse depth estimate of the input image is obtained by the first stream. This initial estimate is

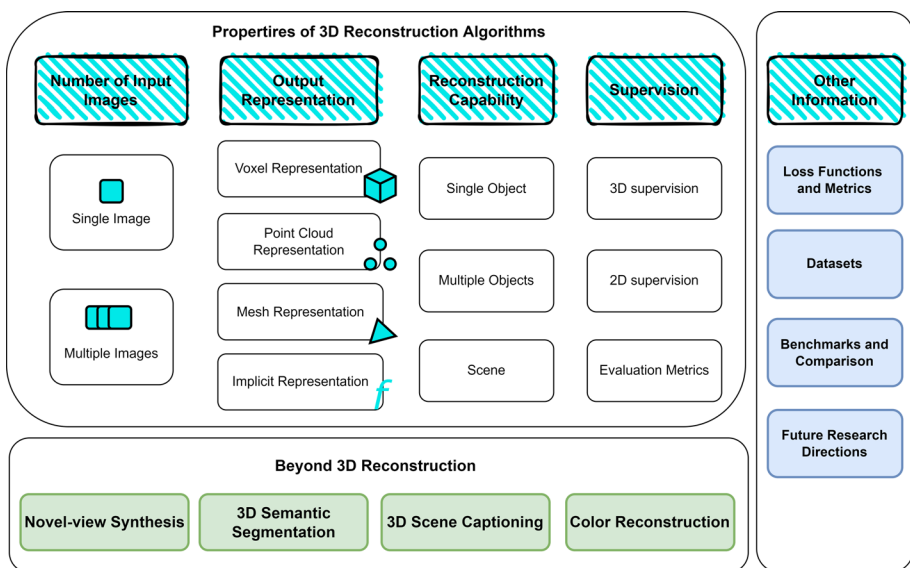


Fig. 1 A visual overview of this research. 3D reconstruction algorithms are categorized and reviewed based on different properties

then fed through the second stream, yielding the final refined depth map. However, the estimated depth map is blurry due to the use of L2 loss in the training process.

Some works first infer multiple depth maps from a single image, and then fuse them to reconstruct 3D shapes of objects (Tatarchenko et al. 2016; Lin et al. 2018). For example, Tatarchenko et al. (2016), proposed an encoder-decoder-based CNN that takes in a single RGB image of an object as well as a desired viewing angle θ to render the object in the desired pose together with its corresponding depth map. 3D reconstruction is made possible by calling the network multiple times, each time with a different θ , and using a post-processing step to re-project the results into a common 3D space and infer the full 3D shape.

Much research have been done on depth estimation, either from single or multiple images. Since 3D reconstruction aims to infer the complete 3D structure of objects, the reader could refer to Bhoi (2019) and Laga et al. (2020) to learn more about depth estimation methods.

3 Cubic world: reconstructing in the voxel format

Occupancy grid: This representation discretizes the three-dimensional space into a grid of uniform cubic parts. Each part has two states: empty or occupied. Due to memory limitations, the grid resolution is often limited to 1024^3 and lower. Despite this limitation, this type of representation easily fits into the deep learning frameworks. Therefore, models that infer 3D shapes using this representation are very easy to implement.

Choy et al. (2016) presented an end-to-end approach for 3D reconstruction based on 3D convolutional and LSTM (Hochreiter and Schmidhuber 1997) networks. Their algorithm, namely “3D-R2N2”, generates the 3D model of an object in the voxel space by receiving one or more images of the object and its bounding box as input. They have proposed two network structures; the first model has no return connections and is more shallow. The second model is deeper and has return connections. Both models have three main parts. These parts include: an encoder, a 3D convolutional LSTM, and a decoder (Fig. 2). The encoder part encodes each input image into a 1024-dimensional vector. This vector is then fed into the 3D convolutional LSTM, which consists of a grid of $4 \times 4 \times 4$ modified LSTM cells. The mentioned LSTM cells do not have output gates, and their hidden states $h(t)$ are computed based on applying convolution to the hidden states of their neighboring cells. Intuitively, this module acts as the model’s memory and considers information from different viewpoints of the object. Finally, the decoder part recovers 3D shape of the object in a 32^3 voxel grid. The low resolution of the output and slow inference time due to the sequential processing of input images can be listed as drawbacks.

To be able to increase voxel space resolution without a cubic growth in resource requirements, Tatarchenko et al. (2017) proposed OGN (Octree Generating Network). They represent 3D models in voxel space using an Octree-based structure (Meagher 1980). The proposed model has an encoder-decoder-based structure. After the encoding stage, the decoder generates a coarse, low-resolution 3D model. The resolution is then increased by some octree-generating layers. Each of these layers divides every voxel into eight parts depending on whether it is occupied or not. If the voxel is empty, the layer does not subdivide it, saving memory. On the other hand, if the input voxel is occupied, it is divided into smaller parts. The network then classifies these smaller voxels as either occupied, mixed,

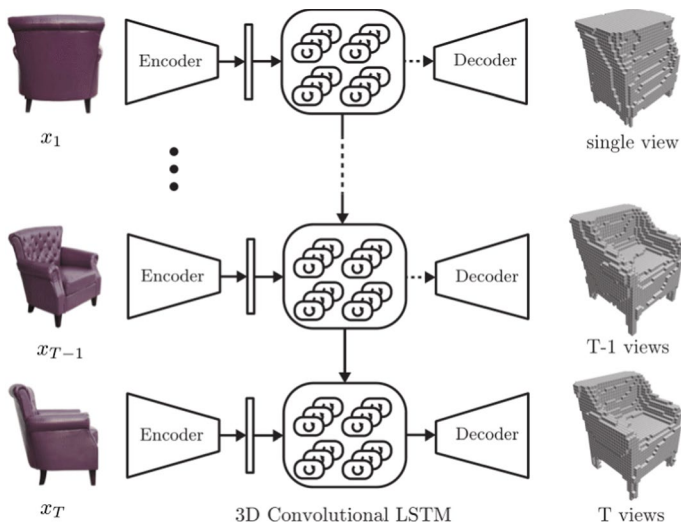


Fig. 2 The model structure in 3D-R2N2 (Choy et al. 2016). An encoder-decoder with a modified convolutional LSTM in between, acting as a memory module to store cross-view information

or empty. The voxels classified as “mixed” are propagated into the next layer for further refinement. The authors have used categorical cross-entropy as the training objective. Figure 3 depicts the learning process in more detail.

In Xie et al. (2019) researchers propose an encoder-decoder-based model for 3D reconstruction that outperforms OGN’s performance by 6% in terms of IoU while having half the number of parameters. A context-aware fusion module takes coarse volumetric predictions from the encoder and fuses the information into one single accurate prediction. However, their method can only reconstruct in 32^3 voxel space, and the performance of the model at higher resolutions has not been investigated. Later, to support higher resolutions (64^3 and 128^3), the authors propose Pix2Vox++ (Xie et al. 2020). The proposed method uses the

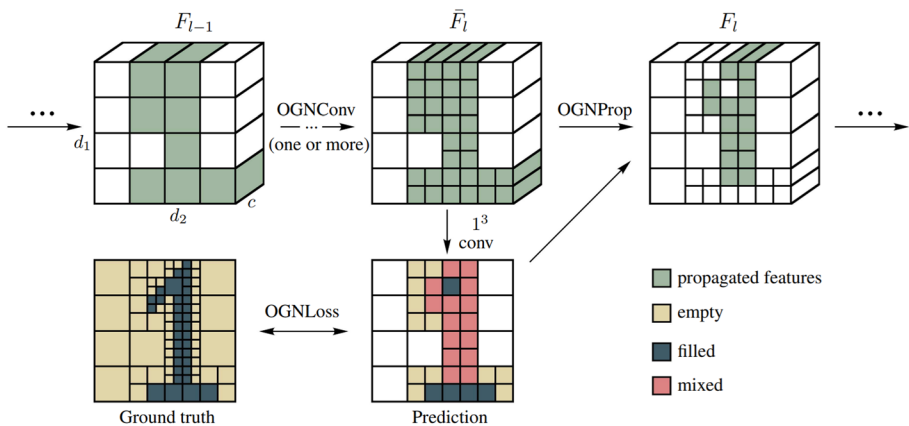


Fig. 3 The learning process of OGN(Tatarchenko et al. 2017) in which a coarse voxel estimate is further refined by division and classification of voxels hierarchically

ResNet (He et al. 2016) backbone in the encoder part instead of the VGG-16 (Simonyan and Zisserman 2014) which reduces the parameter count by 25% and inference time by 5%. Moreover, the overall IoU on the ShapeNet (Chang et al. 2015) dataset has been increased by 1.5%.

Given a single RGBD image of an object as input, 3D-RVP (Zhao et al. 2021b) reconstructs the 3D geometry of an object in two stages. In the first stage, a voxel representation is derived from the input depth image and passed through an encoder-decoder network with skip connections to perform a coarse estimate of the object's shape. With the help of a point-sampling strategy, some points that the model is uncertain about them ($P_{occupancy} \approx 0.5$) are sampled from the voxel grid. The feature maps of the encoder-decoder network are interpolated for each point to obtain point-wise features. A shared MLP named "point-head" predicts the occupation probability of these sampled points using point-wise features. Finally, a fine-grained 3D shape is formed in a voxel grid by combining the 4x up-sampled results of the encoder-decoder network with the predictions of the point-head.

Since the introduction of transformers by Vaswani et al. (2017), they have demonstrated promising results in various tasks of NLP. Recent works in computer vision have focused on incorporating transformers into their models. This was happened after some works achieved state-of-the-art performance in their tasks with the help of transformers, such as ViT (Dosovitskiy et al. 2020), that uses a transformer encoder on a sequence of image patches. Similarly, VoiT (Wang et al. 2021a) aims to reconstruct the 3D shape of an object given multi-view images. This is done by using transformers both for feature encoding and multi-view fusion. The model consists of a 2D-view encoder and a 3D-volume decoder. In the first phase, a shared-weight CNN extracts a set of features from each input view. The resulting vector containing the initial view embeddings passes through a stack of 6 basic encoder blocks, each comprised of a divergence-enhanced view attention and a position-wise feed-forward neural network. The attention layer explores a global relationship among multiple views. In the second phase, the 3D volume decoder learns the global correlation among different spatial locations with the help of a multi-headed volume attention layer. It explores the relationship between the view and spatial domains with the help of the "multi-headed view volume attention" layer. The obtained results from the previous phase pass through a position-wise feed-forward network. A linear function projects the output embeddings of each 3D volume into a 3D output space. Finally, the predicted 3D volumes are reshaped and grouped to form the final reconstructed shape. The method achieves a new state-of-the-art for multi-view 3D reconstruction on ShapeNet (Chang et al. 2015) with 30% fewer parameters than the other recent CNN-based methods. It also has a slightly better scaling capability on the number of input views. For more details about the scaling capability of multi-view models, please refer to Sect. 13, Fig. 26.

Summary: As voxel grids are discrete, the output resolution must be kept at low levels due to resource constraints. Thus, the reconstructed shapes are low-detailed, and the reconstruction accuracy is low. Additionally, this type of representation cannot be directly used in some applications such as game and movie industries, as it requires a different algorithm to obtain more flexible and natural outputs. While most of the reviewed works use 3D CNNs with an encoder-decoder-based structure, 3D-R2N2 employs Convolutional LSTMs to memorize shape information across different viewpoints. However, LSTMs increase the training and inference time due to their sequential nature. Pix2Vox is easy to implement but does not support multi-view inputs. Moreover, the reconstruction resolution is very low. Pix2Vox++ has 4 times higher output resolution and supports multi-view inputs, achieving better accuracy. OCTree mitigates the memory limitations of voxel-based representation by proposing a learned OCTree-generating framework. Therefore, the output

resolution can be as high as 1024×1024 . VoiT utilizes transformers for 3D shape reconstruction and achieves state-of-the-art results with 30% fewer parameters than the other recent CNN-based methods. This algorithm's downside is that the output resolution is still limited to 32×32 .

4 XYZ: point-cloud based 3D reconstruction

A point cloud is simply a set of unordered 3D points in a coordinate system. Compared to ordered structures like meshes that store combinational connectivity patterns, predicting 3D shapes in point-cloud representation is less challenging.

Fan et al. (2017) proposed PSGN that takes a single RGB image of an object and reconstructs its shape by predicting the corresponding 3D point cloud. They provide three models: vanilla, two-branch prediction, and hourglass. Each model includes two main parts: the encoder and the predictor. The encoder performs a set of convolutional operations on the input and yields a feature vector. This vector is then fed into the prediction module, generating a point set of $N = 1024$ 3D points. A random variable is also fed into the algorithm to simulate uncertainty in single image reconstruction during the training stage. For each input image, n outputs are predicted using n different random variables. The distance between the predicted and ground truth point sets is then calculated. Further, the “Min of N ” (MoN) loss ensures that the minimum of n distances is small enough compared to the ground truth. Two distance measures have been used, namely the Chamfer Distance (CD) and the Earth Mover's Distance (EMD). According to qualitative comparison results, the network trained with CD loss performs better at reconstructing details, whereas EMD produces more close-packed results.

In order to produce denser results, Mandikal and Radhakrishnan (2019) proposed “DensePCR”, which is capable of predicting 16x more points. In DensePCR, a multistage model estimates a coarse shape for the input image by predicting a sparse 3D N -point set. These results become denser by applying the “Dense Reconstruction Network.” Each DRN module up-samples the point count by a factor of four. Let X_{p1} represent the points corresponding to the coarse point cloud estimate. The DRN module extracts global features X_g and local features X_l from the 3D N -point set and then concatenates them into a single vector. This vector is input to an MLP to produce a set of $16N$ 3D points, where $N = 1024$. EMD and CD loss functions are used for the first coarse estimate and the next two DRN outputs, respectively. This method outperforms PSGN on ShapeNet, by 5% in terms of CD and by 65% in terms of EMD, respectively.

Lin et al. (2018) proposed a 3D generative framework called “EDPCG”, capable of reconstructing objects in a dense point cloud from a single image. The input image is first encoded into a latent representation using a set of 2D convolutions. The latent representation is then fed into a structure generator network comprised of a stack of 2D transposed convolutional layers. The structure generator module estimates the object's structure from N different viewpoints. Each estimate contains the 3D coordinates of visible parts at each pixel location as well as a binary mask to distinguish between background and object pixels. By using intrinsic camera parameters and the object pose in each estimate, the 3D points from different viewpoints are transformed into canonical coordinates. The network is then optimized on the training data with the help of a differentiable rendering module that synthesizes novel depth images from dense point clouds. According to the authors, this approach for 2D optimization is 100x faster than 3D-based optimization methods such

as Chamfer distance. The value of the loss function is derived by summing up the binary mask loss and the L1 distance of depth values. Figure 4 depicts an overview of this method.

3D-LMNET (Mandikal et al. 2018) illuminates the importance of learning a good prior over 3D point clouds for effective knowledge transfer between 2D and 3D domains for the task of single-view 3D reconstruction. The authors first trained an auto-encoder on 3D point cloud data. The point cloud encoder network learns a latent space for 3D point clouds using Chamfer distance as the training objective. Once the auto-encoder is trained, an image encoder is trained to map the input to this learned embedding space. More precisely, given an input image and its corresponding ground truth point cloud, the image encoder network processes the image to generate a latent vector (Z_I). At the same time, the ground truth point cloud is fed into the frozen 3D auto-encoder to generate the latent target vector (Z_P). The image encoder weights are optimized to minimize the L_2 or L_1 loss between these latent vectors. Additionally, in a second variant, the authors used the idea of Variational Auto Encoder (VAE) (Kingma and Welling 2013) to generate multiple plausible outputs instead of one. Similar to PSGN, this research provides a solution for handling reconstruction uncertainty from a single image. Figure 5 visualizes the training stages.

Zou and Hoiem (2020) propose SGPCR that reconstructs the dense point cloud of an object from a single RGB image. Besides the performance improvement over many point cloud reconstruction methods, the novelty of this approach is that it can reconstruct a partially occluded object by completing its predicted visible silhouette. Before the reconstruction step, a neural network with a U-Net structure is trained to complete the occluded silhouette. The authors have generated a synthetic dataset for network training and silhouette completion. After the completion step, an encoder-decoder network with a ResNet-50 backbone infers the 3D point cloud from the completed silhouette. It first infers a coarse point cloud consisting of 1024 points. A point cloud refinement step up-samples the points by a factor of four. Finally, a post-refinement step fits the surfaces on point clouds, smooths them, and uniformly re-samples points from the smoothed surfaces again. The loss function for the 3D reconstruction network is a combination of different losses, including Chamfer and 2D re-projection. This method achieves a slightly worse CD (1% higher) and a much better (15.2% lower) EMD compared to 3D-LMNET on ShapeNet.

Summary: Methods that use point cloud representation perform better at fixed memory usage than those representing shapes in voxel grids. Among the reviewed works, PSGN has a relatively simple network architecture, and its output resolution is limited. DensePCR reconstructs at a higher resolution, but the training time is high since it applies 3D supervision in multiple stages of the algorithm. 3D-LMNET produces accurate reconstructions;

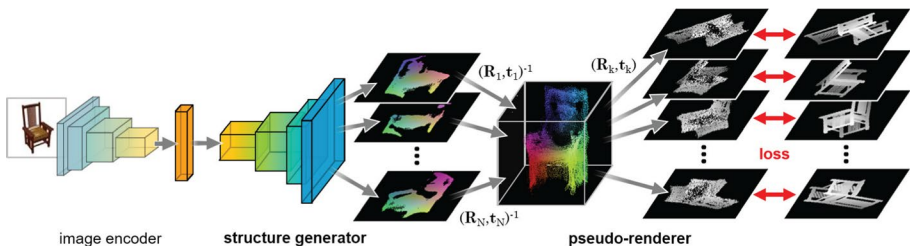


Fig. 4 The model structure and the training procedure in Lin et al. (2018). With the help of a pseudo-renderer, the model is trained with 2D supervision. Experiments have shown a 100× decrease in training time compared to 3D optimization

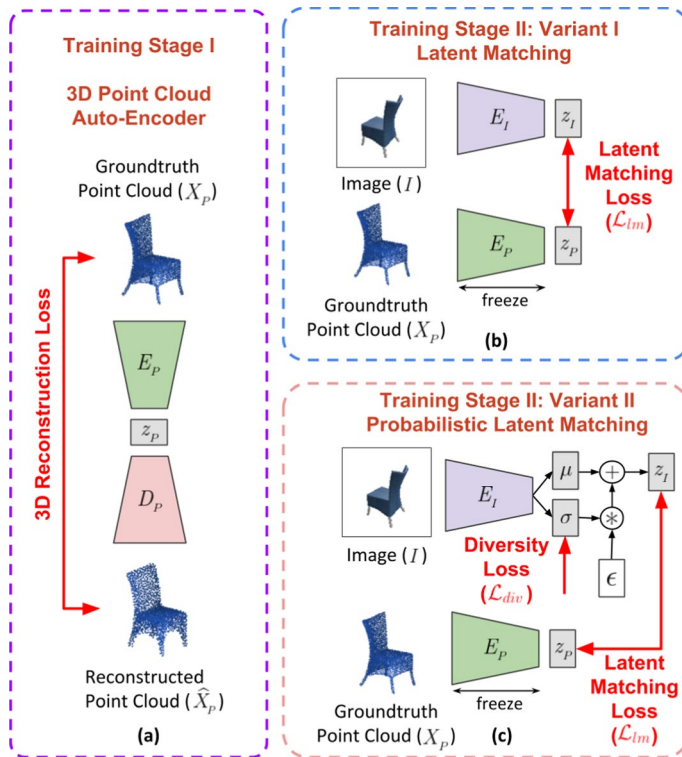


Fig. 5 An overview of the training stages in Mandikal et al. (2018). In the first stage, an auto-encoder is trained by minimizing Chamfer distance. In the second stage, the encoder network is fine-tuned to produce the same latent representation as the target 3D point cloud of the object by minimizing latent matching loss

however, two training stages must be performed, and the decoder's knowledge is limited to the shape categories present in the training data. By applying 2D supervision instead of 3D supervision, EDPCG improves the training time by a factor of 100, but some of the reconstructed shapes lack details. SGPCR applies both 2D and 3D for training and handles occlusions better.

4.1 Point Cloud Completion

Given sparse or partial input observations, the challenge entails retrieving the lost part of an unordered point set. This issue arises due to algorithmic errors, which might be caused by object occlusion, a lack of input data, or a complex object structure. It is worth mentioning that some laser scanners, such as LiDAR, could produce incomplete point clouds for various reasons, including the environmental noise. This emphasizes how critical it is to develop point cloud completion techniques. Since this task does not fully relate to the original task of image-based 3D reconstruction, we only mention a few works.

Huang et al. (2020) proposed PF-Net for point-cloud completion. Their model receives an incomplete point-cloud and tries to complete the input point set in a coarse-to-fine manner. This model consists of three main parts: a triple-scale encoder, a pyramid decoder, and a discriminator network. Figure 6 illustrates the point cloud completion process. The

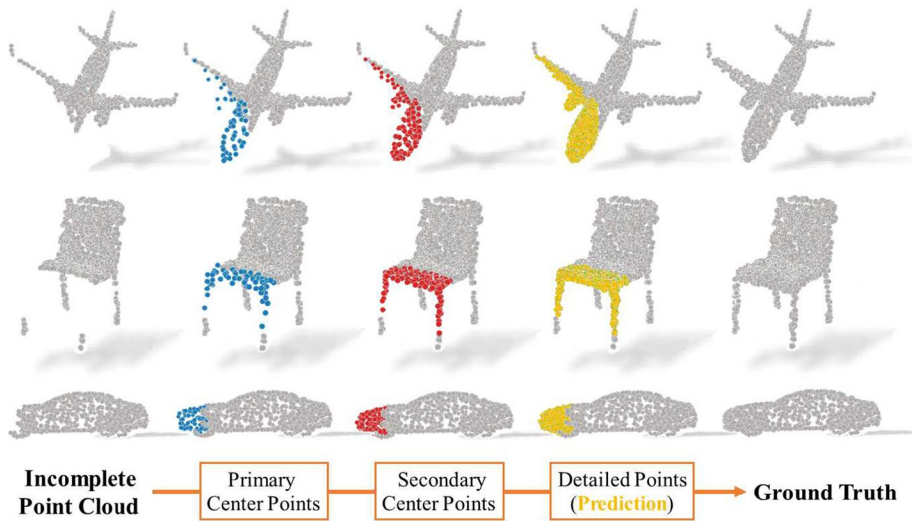


Fig. 6 Illustration of point cloud completion process in Huang et al. (2020)

input point cloud is first sampled on three different scales. To maintain the structure of the 3D model during sampling, the Iterative Farthest Point Sampling (IFPS) (Eldar et al. 1997) algorithm is used. The encoder part extracts the features from these three scales and concatenates them together. From the resulting 1920×3 vector, a 1920×1 vector called V is generated using an MLP. In the decoder part, the vector V is fed into a fully connected network with three layers. Then each of these layers hierarchically predicts the points corresponding to the incomplete part. So that, from the top fully connected layer, a point set with low density and accuracy is obtained. The result of the top FC layer is combined with the middle layer's result. This result is then combined with the bottom layer to create the final result. CD is used as the cost function to compare the results from each level of the pyramid decoder with its reference point cloud. An adversarial discriminator network (Goodfellow et al. 2014) is also trained to increase the accuracy of the model. Figure 7 visualizes the decoding process and the adversarial training procedure. This research proposes an enhanced point encoder compared to PointNet (Qi et al. 2017a). Instead of applying max-pooling to the last feature vector to obtain the final encoded vector, it concatenates multi-level features to form it, capturing both local and global point features.

Pointnet++ (Qi et al. 2017b) was used in research by Zhang et al. (2020) to extract point features while taking local and global features into account with multiple stages of sampling and grouping. Before the completion process, they provide more cues to the model by separating features of known parts and missing parts of the object. The features are then expanded and given to a shared MLP to predict a coarse model, which is further refined using another network.

SnowflakeNet (Xiang et al. 2021) improves upon the reviewed works by utilizing the representational and decoding capabilities of transformers. The feature extractor uses three layers of set abstraction from PointNet++ along with a point transformer model (Zhao et al. 2021a) to encode the incomplete input shape into a shape code. A seed generation module is designed to produce a coarse but complete point set that maintains the geometry and structure of the target shape. More detailed, the module extracts point-wise features

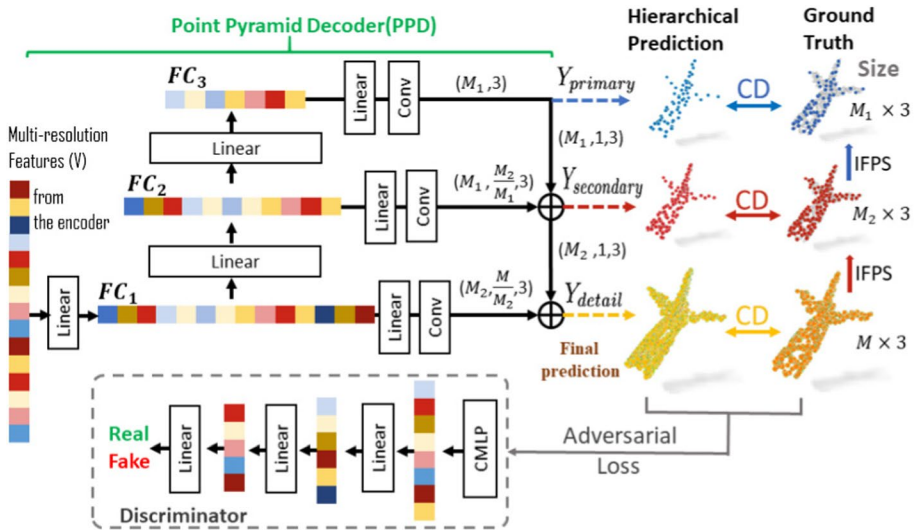


Fig. 7 The decoder structure in Huang et al. (2020). After the encoding phase, the multi-resolution features (V) are fed to a 3-layer MLP to predict the completed point cloud hierarchically. A discriminator network is responsible for improving prediction quality

that capture both missing and known parts of the shape. The module then infers a coarse point set for the shape that is concatenated with the original points from the input. Farthest Point Sampling (FPS) (Qi et al. 2017b) is applied to obtain the final seed shape at a specified resolution. A novel Snowflake Point Deconvolution (SPD) module is proposed to progressively increase the number of points in three steps. Child points are generated from the parent points of the input in such a way that each child inherits its parent's point features. To facilitate consecutive SPDs to split points coherently, a skip transformer is proposed to capture the shape context and the spatial relationship between parent points and their child points.

There are many other approaches published to solve the shape completion task; among them are Huang et al. (2021), which focuses on reducing network parameters and inference time while achieving promising results by introducing a novel recurrent forward network for point cloud completion, and Sarmad et al. (2019), which utilizes reinforcement learning.

5 Surface based 3D reconstruction

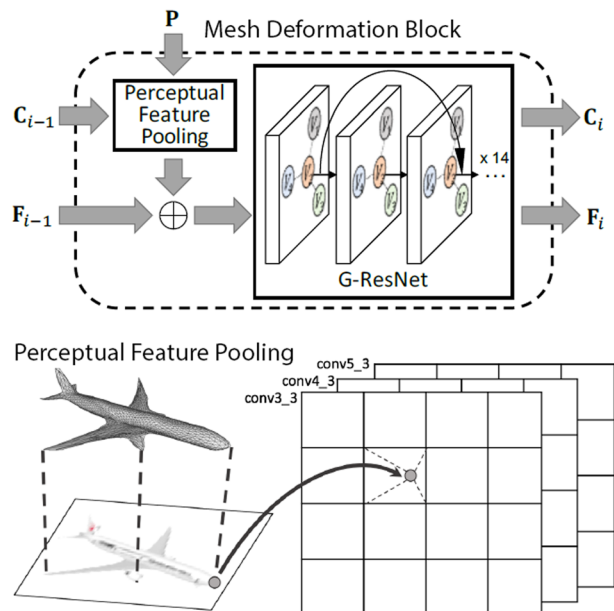
Formerly reviewed representations cannot be directly used in some applications, e.g., animation and cinema. Surface-based representations have the advantage of being ready to be directly used in various applications. These representations are uniformly deformable and consume less memory than voxel-based representations, as they only model the surface. However, such representations do not easily fit into deep learning frameworks.

Wang et al. (2018a) proposed Pixel2Mesh which takes a single 2D image of an object as input and infers the corresponding 3D mesh by utilizing Graph Convolutional Networks (GCN) (Scarselli et al. 2008; Bronstein et al. 2017; Defferrard et al. 2016). The proposed model deforms an initial ellipsoid mesh in three stages to reconstruct the object. The initial

ellipsoid is first fed into a mesh deformation block (Fig. 8), which deforms the input mesh using early perceptual features of the input image. In each of the next two stages, after applying edge-based up-sampling to the mesh model, the up-sampled mesh is fed into a deformation block for further refinement. The mesh deformation block refines the up-sampled results of the previous stage by projecting the vertices of the 3D mesh model into 2D space. It then assigns each vertex a feature vector by aligning 2D vertex locations to corresponding perceptual features extracted from the input image. It then infers new locations for each vertex by applying a set of graph convolutional blocks to these features. In terms of the loss function, the researchers used Chamfer loss along with normal loss. The normal loss function is defined in Sect. 11 as Eq. 8. It forces each target surface normal to be perpendicular to the nearest predicted edges. To solve the convergence problem of the network, which tends to get stuck at local minima, two regularization terms, namely Laplacian and edge-length, were added to the loss function. The ablation study shows that edge-length regularization and normal loss have the highest impact on reconstruction quality. One limitation of this algorithm is that it can only receive a single image as input and cannot perform multi-view image reconstruction. Thus, it makes it difficult for the model to estimate the 3D structure of occluded parts of the object. Also, this model cannot reconstruct scenes with multiple objects as it is trained for single-object reconstruction. These limitations prompted the researchers to develop a new model.

A year later, Pixel2Mesh++ (Wen et al. 2019) was introduced. It receives multiple images captured from different viewpoints to reconstruct the object. The algorithm first estimates a coarse 3D mesh model of the object using a previously trained Pixel2Mesh model. This estimate is then fed to a multi-view deformation network. This network first generates several hypotheses for each vertex of the coarse 3D model. Each hypothesis is a possible new location for a given vertex with an assigned probability. After forming a hypothesis graph for each vertex, a graph CNN predicts the vertex movements. In the next step, a feature vector is assigned to each hypothesis, just like in Pixel2Mesh. This is done

Fig. 8 (Top) A mesh deformation block. This block deforms the up-sampled mesh to produce a refined 3D model with finer details by utilizing GCN (Scarselli et al. 2008). (Bottom) Depiction of the perceptual feature pooling process (Wang et al. 2018a)



by projecting the coarse 3D model onto the input image feature maps and extracting the corresponding features for each vertex. The only difference here is that the multi-view features of the object must be handled. The problem with the concatenation of multi-view features is that the feature vector length is not constant and increases with the number of input images. To solve the issue, for each hypothesis, vectors of mean, maximum, and variance of multi-view features are concatenated together in a fixed-size vector. In the next step, the deformation reasoning block assigns a new location for each vertex. This block assigns a weight to each hypothesis and passes it through a softmax function. The final location of the vertex is the weighted sum of its hypotheses.

While the previous methods for 3D mesh reconstruction only learned the displacements of a template mesh to deform it into the target mesh, Pan et al. (2019) introduce a novel topology modification module to prune the faces that deviate significantly from the ground truth. To prune these errors, the network must estimate errors correctly. Therefore, an error estimation network is trained by a quadratic loss to regress the reconstruction errors. Together with a mesh deformation module, the proposed method can reconstruct complex typologies from a genus-0 template mesh at high resolution. Furthermore, a boundary refinement network is also responsible for refining the boundary conditions to improve the quality of the reconstructed mesh. Figure 9 provides an overview of the whole pipeline. The quantitative results are reported for five classes of ShapeNet. These results demonstrate a 17% improvement in terms of CD and a 13.7% improvement in EMD over Pixel2Mesh.

To this end, despite achieving high geometric accuracy, the reviewed methods for mesh reconstruction produce models with self-intersecting meshes. Neural Mesh Flow (NMF) (Gupta and Chandraker 2020) focuses on inferring 3D meshes with high manifoldness. Instead of deforming a template mesh using GCNs (Defferrard et al. 2016), the deformation is performed by learning a diffeomorphic flow from a genus-0 template mesh to a target mesh. Diffeomorphic flows are unique and preserve orientation; therefore, the manifoldness of the input template mesh is preserved after deformation. The authors model the diffeomorphic flows using neural ODEs (Chen et al. 2018). Moreover, to empower the representation of multiple shape categories, the authors stack three layers of neural ODEs to estimate deformation flows gradually. They also apply an instance normalization layer to both the input and hidden features. The Chamfer loss function is applied to all three deformation stages to train the network.

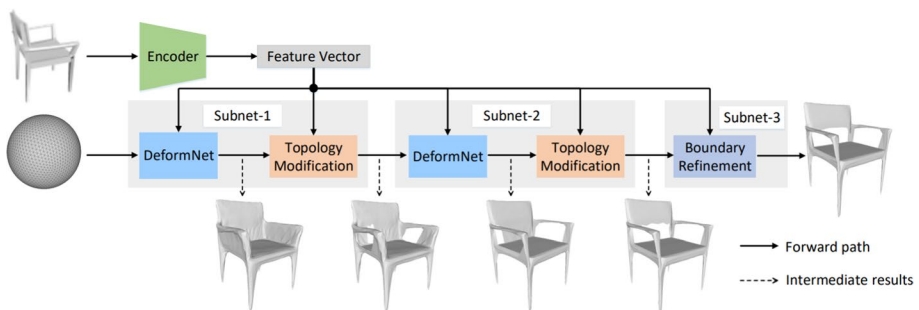


Fig. 9 Overview of the reconstruction pipeline in Pan et al. (2019). An initial Genus-0 template mesh is progressively deformed in multiple stages to produce the final shape. A topology modification module regresses the error regions at each stage and removes them

Summary: Surface-based representations are deformable and require fewer resources than voxel-based representations. Pixel2Mesh was one of the first to leverage graph CNNs to estimate an object's 3D shape in a mesh representation. Pixel2Mesh++ has improved reconstruction accuracy and supports multi-view inputs. Pan et al. (2019) introduce a learned error pruning network to remove faces that deviate significantly from the ground truth. The presence of non-closed meshes and the lack of multi-view input support can be listed as drawbacks of the former method. NMF generates high manifold meshes by learning a diffeomorphic flow from a genus-0 template mesh. However, the resulting meshes are over-smoothed.

6 3D reconstruction using implicit representation

Implicit neural representation is a novel way to parameterize different kinds of signals. Conventional signal representations are usually discrete; for instance, 3D shapes are parameterized as grids of voxels, point clouds, or meshes. In contrast, implicit neural representations parameterize a signal as a continuous function that maps the domain of the signal (such as a 3D coordinate) to whatever is at that coordinate (occupancy probability, Signed Distance Function (SDF) value). Implicit neural representations approximate that function via an MLP network. This MLP, with sufficient layers and hidden units, can approximate any function with arbitrary precision. These methods have the advantage of memory efficiency, enabling the network to infer complex 3D shapes more accurately at arbitrary resolutions without any concerns about resource limitations.

ONet (Mescheder et al. 2019) uses a neural network to approximate the occupancy function of a 3D object denoted as $\phi : \mathbb{R}^3 \rightarrow \{0, 1\}$. The network takes in a query point $q \in \mathbb{R}^3$ conditioned on another input vector $x \in \mathcal{X}$ and predicts the occupancy probability of that point as $p \in \mathbb{R}$. To produce the aforementioned vector x , the authors utilized a ResNet18-based (He et al. 2016) encoder to encode the input image. They further condition the query point on encoded features using conditional batch normalization (De Vries et al. 2017). The training is done by sampling K points from each training sample and predicting the occupancy probability corresponding to each query point. By minimizing the cross-entropy classification loss, the network learns to infer 3D shapes from a single input image. During the inference process, the volumetric space is first discretized into a 32^3 grid of points, and the network predicts the occupancy of these points based on thresholding the occupancy probabilities. After a coarse 3D shape is inferred, the points in the occupied area are up-sampled by incrementally building an octree. The network then evaluates each of these newly generated points to determine their occupancy states. This mechanism is called Multi-resolution Iso-Surface Extraction (MISE), which is illustrated in Fig. 10.

IM-NET (Chen and Zhang 2019) takes in a feature vector extracted by a shape encoder as well as a 3D or 2D point coordinate, returning a value indicating the status of the point relative to the shape (inside or outside). The decoder structure is shown in Fig. 11. The encoder can be either a CNN or PointNet. In other words, the network is trained to solve a binary classification problem by learning a mapping function $f_\theta(p)$, which maps $p \in [0, 1]^3$ to a binary occupancy value. The network is capable of single-view 3D reconstruction. The authors first trained an auto-encoder on the training set to produce a ground truth feature representation. Having obtained these ground truth feature representations, they fine-tuned a pre-trained ResNet (He et al. 2016) encoder to minimize the mean squared loss between the predicted feature vectors. This method performed better than training the

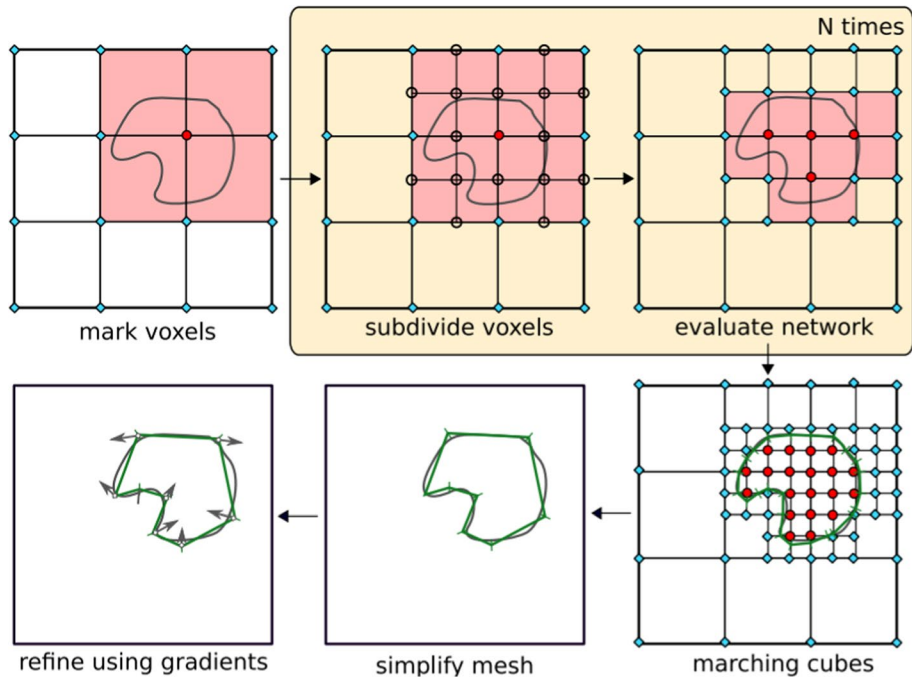


Fig. 10 Inference procedure in ONet (Mescheder et al. 2019)

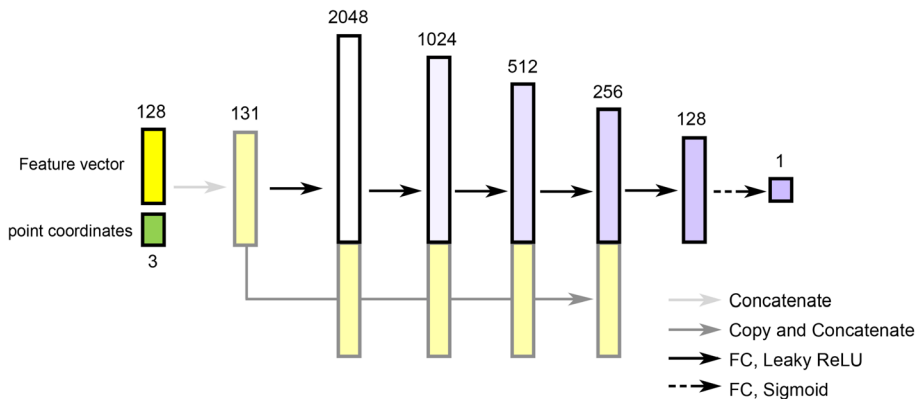


Fig. 11 The decoder structure of IM-Net (Chen and Zhang 2019). This network predicts a query point's inside/outside status from the encoded features. It is also noted that skip-connections help boost the learning process

image-to-shape translator directly. As stated by the authors, pre-trained encoders provide strong priors that can reduce single-view reconstruction ambiguity and shorten the training time by being trained on unambiguous data in the auto-encoder phase. A major limitation of this approach is that the output results have low-frequency errors (e.g., global shape characteristics such as thickness or thinness), which seem to be mitigated by regressing the SDF value of each input point instead of predicting the binary occupancy value.

Deep SDF (Park et al. 2019) implicitly represents the zero iso-surface of 3D shapes as a decision boundary of a feed-forward neural network. The network takes in a latent code Z along with a query point in 3D space and regresses the corresponding SDF value conditioned on the shape code. Points with $SDF = 0$ implicitly represent the iso-surface of the object, which can be rendered through ray casting or rasterization of a mesh obtained with, for example, Marching Cubes (Lorensen and Cline 1987). In order to reconstruct fine details of the shape, the network is forced to focus its prediction near the object's surface (the zero-level set). To achieve this, the predicted SDF values (f_θ) and target values (S) are clipped with a small threshold δ . The clipped values are then used to calculate the L1 loss. This loss function is shown in Eq. (1).

$$\mathcal{L}(f_\theta, s) = \|\text{clamp}(f_\theta(x), \delta) - \text{clamp}(s, \delta)\| \quad (1)$$

where $\text{clamp}(x, \delta) := \min(\delta, \max(-\delta, x))$. The authors also claimed that since the trained encoder is unused at test time, it is unclear whether using the encoder is the most effective use of computational resources during training. So they proposed an auto-decoder instead of an auto-encoder. At the beginning of training, a random initial code is assigned to each shape. The auto-encoder then learns the optimal latent code jointly with the decoder weights during training. At training time, they maximize the joint log posterior over all training shapes with respect to the individual shape codes $\{z_i\}_{i=1}^N$ and the network parameters θ :

$$\underset{\theta, \{z_i\}_{i=1}^N}{\operatorname{argmin}} \sum_{i=1}^N \left(\sum_{j=1}^K \mathcal{L}(f_\theta(z_i, x_j), s_j) + \frac{1}{\sigma^2} \|z_i\|_2^2 \right) \quad (2)$$

Summary: A surface can be implicitly modeled as the zero-level set of a function that is learned by an MLP. The implicit representations have high memory efficiency; therefore, they can represent shapes at high resolutions. However, the inference time of these approaches is considerable, especially at higher resolutions. Another limitation of the reviewed works is that they only accept single-view inputs and do not consider a pipeline for multi-view inputs.

6.1 Implicit 3D reconstruction of clothed human body

Clothed human 3D reconstruction has many applications in virtual reality and medical imaging. Recently, numerous studies have been proposed on clothed human body reconstruction that use an efficient implicit function to model the object's surface. In the following, some of the latest works are reviewed.

Given a single or multiple background-extracted color images of a person, PIFu (Saito et al. 2019) infers a textured 3D surface model of that person by using a pixel-aligned implicit function that is learned by an MLP. Having extracted the input image features, to get the corresponding features for each query point in the 3D space named x , its projection on the input 2D image is obtained. These features, accompanied by the depth value of the query point with respect to the camera, are then fed into an MLP to predict the inside-outside probability of that point. As shown in 12, two different streams have been proposed, one for surface reconstruction and another for texture inference. The latter's network structure is the same as the former, differing only in the output, which is RGB values for each query point.

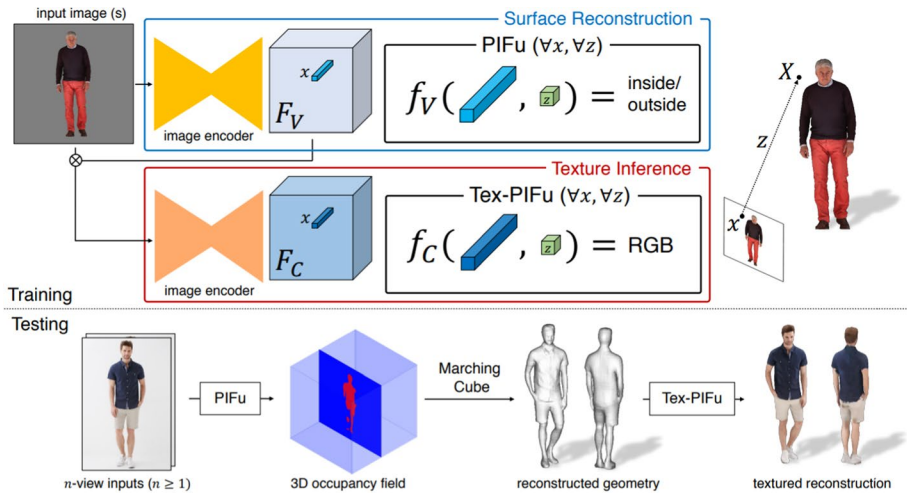


Fig. 12 Overview of PIFu pipeline: Given an input image, a pixel-aligned implicit function (PIFu) predicts the continuous inside/outside probability field of a clothed human. Similarly, PIFu for texture inference (Tex-PIFu) infers an RGB value at given 3D positions of the surface geometry with arbitrary topology (Saito et al. 2019)

Geo-PIFu(He et al. 2020) does not provide a texture inference stream; instead, it adds a new stream to extract latent 3D features and predicts a coarse volumetric shape. For each query point in 3D space, the tri-linearly interpolated latent 3D features are concatenated with 2D latent features extracted by a 2D UNet. The resulting geometry and pixel-aligned features are then fed into an MLP to predict occupancy status. Zheng et al. (2021) have a similar framework to Geo-PIFu, but use the encoded features of an inferred parametric model (Skinned Multi-Person Linear Model, or SMPL) from the input image in its geometric feature composition phase. A body reference optimization step is proposed to prevent the network output from being dependent on the initial SMPL estimation.

PIFuHD (Saito et al. 2020) improves upon PIFu by increasing the feature resolution of the backbone output. They argue that while the implicit representation can represent 3D geometry at any arbitrary resolution, the expressiveness of the features can be further improved. The researchers also add another input requirement, which is normal maps of the front and back, to mitigate the ill-posed problem of single-view reconstruction. These normal maps are produced by pix2pixHD (Wang et al. 2018b) and are fed as features into the pixel-aligned predictors. As shown in 13, the proposed model has two branches: one for the coarse prediction, whose output is used in the second branch, which has higher input, feature, and output resolutions.

In summary, the proposed approaches for human reconstruction using implicit representations are capable of high-resolution reconstruction since the input query points are continuous. Among the reviewed works, PIFu has the least parameter count (about 15 million). Although it fails at reconstructing the back of the object accurately, it still produces promising results. Geo-PIFu has twice the parameters as PIFu and uses geometry-aware 3D features along with pixel features. Besides, considering that supervision is also applied to the rough volumetric shape, the dimensionality and quality of the observation space of the MLP network are increased. The quantitative and qualitative results support the former statement. However, the 3D stream in this model adds a considerable amount of memory

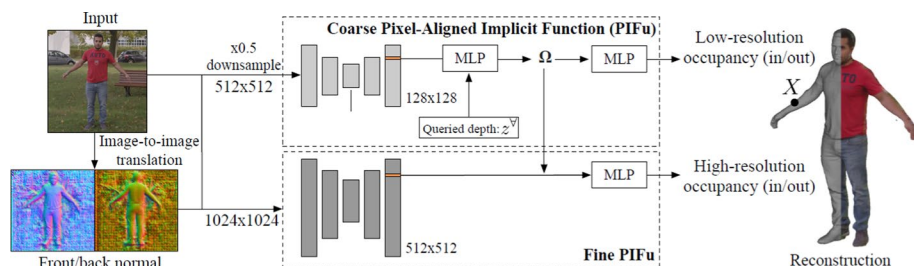


Fig. 13 Overview of Saito et al. (2020). Two pixel-aligned branches contribute to producing a high-resolution 3D reconstruction. The coarse branch (top) captures the global 3D structure, while the fine branch adds high-resolution details to the shape

requirement and runtime to the model compared to PIFu. PIFuHD manages to reconstruct the back of the object better and also recovers finer details. However, the model parameter size is gigantic, with 387 million parameters (about 25 times more than PIFu). Research (Zheng et al. 2021) handles challenging poses well but requires an SMPL model as input, which in many real-life applications is not available.

7 Multiple object reconstruction

All the previously reviewed works were able to reconstruct the shape of a single object. Real-world tasks such as robotic navigation or manipulation and 3D scene understanding require inferring the 3D shape of multiple objects or even an entire scene. An overview of several studies that aim to reconstruct multiple objects follows.

3D-RCNN (Kundu et al. 2018) recovers 3D shapes and poses of all present objects in a scene by receiving a single 2D image with a set of associated object bounding boxes. By learning a small parametric model for each object category, the method uses prior knowledge about the objects' classes and their shapes to infer the 3D structure of objects. The proposed model is trained using a cost function called "Render and Compare". After the forward pass, in which the model infers a 3D shape, the corresponding depth map and semantic segmentation are obtained using the known internal and external camera parameters and then compared with the ground truth. Figure 14 shows different parts of the model in detail. A drawback of this algorithm is that it uses limited 3D shape priors. Therefore, the model can only reconstruct the shape categories present in the training data. Furthermore, as the method processes objects sequentially, the inference time for reconstructing a scene with multiple objects is relatively high.

Mesh-RCNN (Gkioxari et al. 2019) is a multitask algorithm that, given a single 2D image, first detects the objects along with their category. After drawing bounding boxes around each object, it generates its associated 3D mesh model. To infer the 3D shape, it first estimates a coarse voxel-based shape and then converts it to mesh using the Cubify (Gkioxari et al. 2019) algorithm. Figure 15 shows this process. Several graph convolutional layers have been used to refine the predicted mesh further. In the training stage, binary cross-entropy and Chamfer loss functions have been used for voxels and meshes, respectively.

Popov et al. (2020) proposed "CoReNet" that is capable of reconstructing multiple objects from a single image, each in their original pose, as opposed to many works that

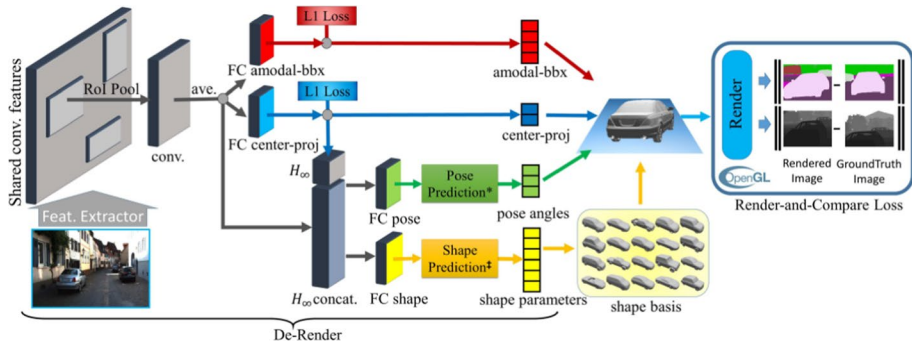


Fig. 14 Overview of 3D-RCNN (Kundu et al. 2018). After extracting RoIs, each of them passes through three streams sequentially. These streams include: amodal-bounding box regression, object canonical center prediction, and pose and shape prediction. The “Render and Compare” method helps supervise the training procedure in 2D

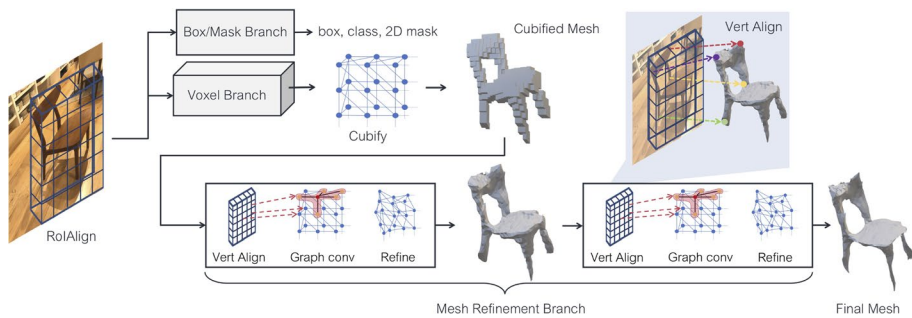


Fig. 15 An overview of the Mesh-RCNN pipeline. The proposed algorithm localizes objects in the input image and predicts their shape category. In the next phase, it infers a coarse voxel-based shape for each object. It then converts the shape into mesh representation and refines it by applying a set of graph convolutional operations. Gkioxari et al. (2019)

reconstruct objects in their canonical pose (Choy et al. 2016; Wang et al. 2018a; Fan et al. 2017). The model, just like the majority of previous works, has an encoder-decoder architecture but adds ray-traced skip connections between the encoder and decoder networks. The skip-connections allow the local 2D information to propagate into 3D space in a correct physical manner. Figure 16 shows the model architecture and ray-traced skip connections. The researchers introduce a new hybrid output representation that combines the best of both voxel grids and implicit representations. The output is a $W \times H \times D$ grid of points, distanced v from each other. The location of these grid points can be changed with an offset of $\bar{o}$, smaller than v . To infer the 3D shape of an object at any desired resolution; the algorithm is repeatedly called with different grid offset values. This helps keep the memory usage constant while reconstructing fine details. In each call, the query points of the grid are also fed into the network to produce the occupancy probability of them over C different classes. The value for C represents the maximum number of objects present in the scene plus one additional class for background. This helps the network handle object occlusions while reconstructing multiple objects. The researchers argue that categorical cross-entropy

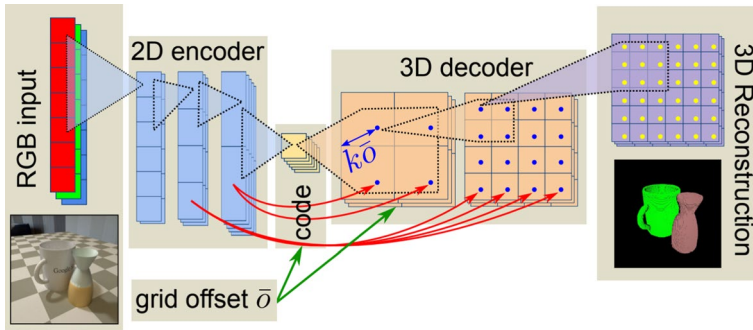


Fig. 16 CoReNet model architecture. The skip connections between the encoder and decoder parts trace the rays from the encoder to the corresponding elements in each decoder layer. The decoder grid offset enables high-resolution reconstruction. This is done by calling the network multiple times with different offset values (Popov et al. 2020)

as the training objective leads to high sparsity of output as most grid points remain unoccupied. Hence, they introduced a new training objective based on IoU, which supports continuous values and multiple classes. The loss function is formulated as:

$$IoU_g(y, \hat{y}) = \frac{\sum_{p \in G} \sum_{c=1}^{C-1} \min(y_{pc}, \hat{y}_{pc}) \mu_{y_{pc}}}{\sum_{p \in G} \sum_{c=1}^{C-1} \max(y_{pc}, \hat{y}_{pc}) \mu_{y_{pc}}}, \quad (3)$$

$$\mu_{y_{pc}} = \begin{cases} 1 & \text{if } y_{pc} = 1 \\ \frac{1}{C-1} & \text{if } y_{pc} = 0 \end{cases}$$

Where y_{pc} and $\hat{y}_{pc}$ are ground-truth and predicted occupancy probability of point p belonging to class c , respectively. G denotes the set of grid points. Since in the ground truth one-hot encoding, $C - 1$ values are zero, $\mu_{y_{pc}}$ balances this sparsity.

Engelmann et al. (2021) propose a method for real-time multiple-object 3D reconstruction from a single image. Based on the idea of CenterNet (Zhou et al. 2019), the method first detects objects as points in a scene. For each detected object center, the network first retrieves the object's class from its prior knowledge from the training data. Afterward, the network predicts each object's 9-DoF (sRT) bounding boxes. With the help of a collision loss, the reconstruction results are made physically plausible without intersections between objects. The absolute 3D IoU score on ShapeNet is not better than the previous work "CoReNet". However, the reported mIoU on Pix3D (Sun et al. 2018) shows a slight improvement over "CoReNet".

Shin et al. (2019) with a single RGB image as input, reconstruct the whole 3D structure of a scene in real-time using a fully convolutional neural network. To represent a 3D shape, the authors used a multi-layer depth representation. This representation consumes less memory than a voxel-based representation, allowing reconstruction at higher resolutions. Unlike a group of algorithms such as 3D-R2N2 that perform object-centered prediction, this method infers the 3D scene structure in a viewer-centered manner. This has been shown to improve the generalization of the algorithm (Shin et al. 2018). After extracting the features from a 2D input image, a 5-layer depth map ($D = \{D_i | i = 1, 2, \dots, 5\}$) as well as the semantic segmentation of depth layers 1 and 3, are estimated using an encoder-decoder-based network structure with skip connections. The structure of the semantic segmentation

network is the same as the depth prediction network, except that its output has 80 channels (40 different object classes per depth map). Since the depths of unseen parts of objects are not yet estimated, the researchers introduce the “Epipolar Feature Transformer” (EFT) network to estimate the full 3D structure of the scene. The 2.5D predictions from the previous stage, along with the feature maps, are fed into the EFT network. These predictions are used to estimate the scene’s depth and semantic segmentation from a new virtual view. In cases where there are many objects in the scene, a virtual overhead view is used. This means that a virtual camera is positioned above and in the center of the scene. The EFT network estimates the features for the new virtual view by appropriately transforming the original input image features. After the prediction of the multi-layered depth maps from two different viewpoints, the 3D structure of the scene is estimated in the form of triangular meshes or voxels. To achieve this, the researchers considered an empty occupancy grid with a high spatial resolution for the scene. Each voxel is then projected onto the camera plane. According to the depth map, if the depth d of a pixel falls in a certain range ($D1 < d < D2$ or $D3 < d < D4$), the corresponding cell in the voxel grid will be considered occupied. The 3D reconstruction accuracy of this method is about twice as high as previous algorithms such as Tulsiani et al. (2018).

Murez et al. (2020) developed an algorithm called ATLAS for 3D scene reconstruction that infers the 3D mesh model of an entire scene and semantic segmentation of present objects. It is all done by receiving multiple posed images of the scene. In this algorithm, 2D images, camera parameters, and poses are given as input. A CNN then extracts the features of the images. The corresponding features of each pixel in the input 2D image are then projected onto each voxel along the ray in a reference occupancy grid. Multiple feature vectors are generated for corresponding voxels in the case of input images having overlaps. To have a fixed-size feature vector for each voxel in the reference grid, the authors calculate the weighted moving average of these vectors. After this step, an encoder-decoder-based CNN network is utilized to predict Truncated-SDF values for the occupancy grid. The advantages of this approach include end-to-end training and the multi-tasking capability of the model. As mentioned before, in addition to the 3D mesh model, semantic segmentation of 3D objects is also predicted. However, its segmentation performance is much lower than the models specifically trained for this task. For instance, the mean IoU obtained on the ScanNet (Dai et al. 2017) dataset in the task of 3D semantic segmentation is 34%. While the MinkowskiNet (Choy et al. 2019) achieved 73.4%.

There are memory and time efficiency issues with methods like (Murez et al. 2020), which infer multiple volumes for all input views and then aggregate them. Moreover, processing a huge 3D feature volume by applying 3D CNNs leads a slow runtime of the algorithm. NeuralRecon (Sun et al. 2021) improves runtime by a factor of ten compared to ATLAS while achieving a slightly better F-Score on the ScanNet dataset. Given a monocular video as input, NeuralRecon processes local fragments of the video to reconstruct the entire scene sequentially. A set of N keyframes is extracted from each local fragment. The process of predicting local TSDF volumes is done in a three-level, coarse-to-fine manner. In this process, after the extraction of multi-level features from the keyframes, the results are then back-projected into three feature volumes corresponding to each scale. An MLP is responsible for predicting the occupancy probability and TSDF values of the voxels at each scale. By using implicit representations, the memory efficiency of the algorithm is increased. A convolutional variant of GRU is responsible for making reconstruction consistent between fragments. The whole pipeline is depicted in Fig. 17.

Summary: In this section, some multi-object and scene reconstruction techniques were reviewed. Starting with 3D-RCNN, it uses prior knowledge to reconstruct voxel-based 3D

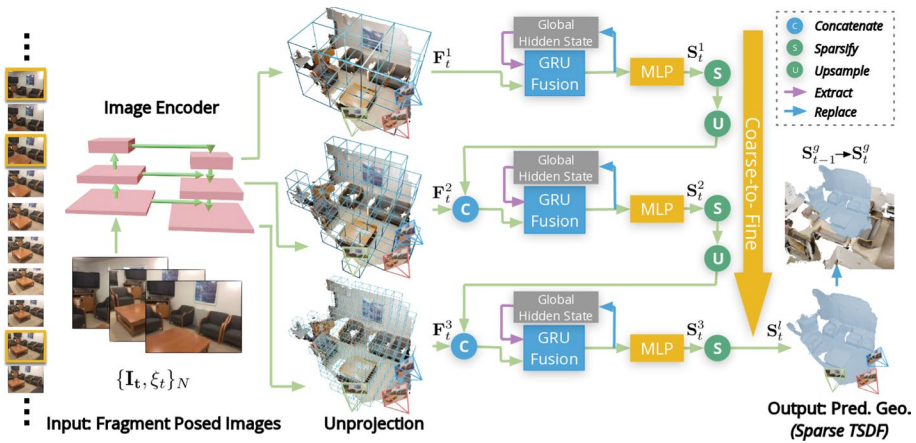


Fig. 17 The proposed pipeline for 3D reconstruction of the whole scene in Sun et al. (2021)

shapes given a single image. However, it needs the corresponding object bounding boxes as an additional input. Moreover, the model's reconstruction coverage is bound to seen object categories during training, and the inference time is high due to the sequential processing of present objects. On the other hand, Mesh-RCNN can reconstruct in a mesh representation which is more resource efficient and ready to use in real-life applications. Nevertheless, object detection errors affect reconstruction accuracy. Most previous works have encoder-decoder architectures, while CoReNet adds ray-traced skip connections between encoder and decoder networks. It reconstructs each shape in each original pose and uses a hybrid representation that combines the advantages of both voxel grids and implicit representations. Despite the novelties of this work, the inference time is still high. Researchers in Points2Objects (Engelmann et al. 2021) follow similar steps as in Mesh-RCNN; but use a point-based object detector and propose a novel collision loss that makes the results physically plausible without intersections between objects. Among the reviewed scene reconstruction methods, Shin et al. (2019) use multi-layered depth representation as the output representation, requiring fewer resources. However, the method's capability is limited to single-image reconstruction. ATLAS and NeuralRecon both reconstruct 3D scenes by receiving multiple posed images of the scene as input. NeuralRecon proposes a more efficient method than ATLAS, using implicit surface representations and GRUs to perform cross-fragment aggregation. Therefore, it is more memory efficient than ATLAS and can run in real-time.

8 Multi-view stereo

Multi-view stereo (MVS) is the general term given to a group of techniques that use stereo correspondence as their primary cue and use more than two images (Furukawa et al. 2015). In other words, "multi-view stereo" refers to the task of reconstructing a 3D shape from calibrated overlapping images captured from different viewpoints (Sinha 2014). An example of the MVS pipeline is depicted in 18. Various representations can be used in such algorithms, depending on the application. Most of the learning-based methods use depth maps or volumetric representations. Compared to depth-map-dependent methods,

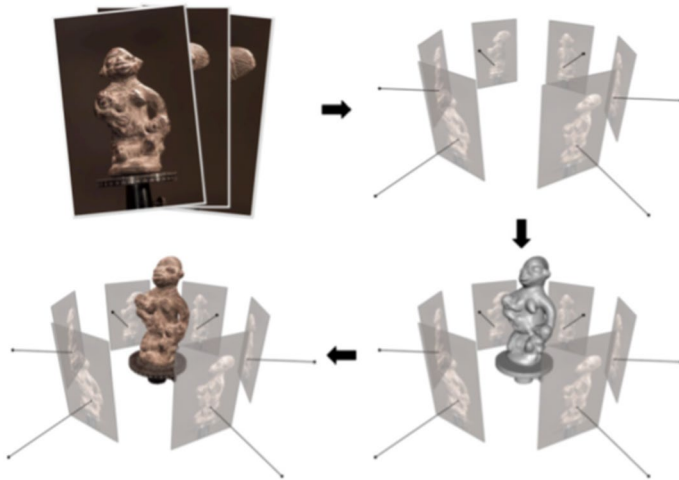


Fig. 18 Example of a multi-view stereo pipeline. Clockwise: input imagery, posed imagery, reconstructed 3D geometry, textured 3D geometry (Furukawa et al. 2015)

volumetric representations have the disadvantage of space discretization errors and lower output resolution due to memory limitations. Based on the latest benchmarks (Knapitsch et al. 2017; Aanæs et al. 2016), depth-map-based approaches offer state-of-the-art results. The proposed methods are mostly based on plane-sweep stereo (Collins 1996), which form a cost volume out of warped multi-view features, regularize them and then estimate the depth. In the following, some notable works are reviewed.

One of the first approaches to learning-based MVS is SurfaceNet (Ji et al. 2017), which takes a set of images as input with corresponding camera parameters and outputs the voxel surface reconstruction. It first pre-computes a representation called Colored Voxel Cube (CVC) for each of the views. This is done by projecting that view's pixels onto a 3D voxelized grid so that each voxel in CVC has a color value. It then uses a 3D CNN to regularize and infer the surface voxels.

MVSNet (Yao et al. 2018) performs 3D reconstruction by first extracting 2D features from a reference image and source images. The extracted features of source images are each warped onto multiple fronto-parallel planes of the reference camera frustum, forming N feature volumes, where N is the number of views. These volumes are then aggregated into a single cost volume by computing the variance of values across cost volumes of different views. As a previous step to forming the probability volume, 3D CNNs help regularize the cost volume. The volume is then converted to a depth map by computing soft argmin over probability values of different depth levels for each pixel. Soft argmin is preferred over the argmax operation due to its differentiability and ability to produce sub-pixel estimations. The estimated depth map is refined further by filtering outlier depth predictions. Point cloud 3D reconstruction of the scene is obtained by fusing depth maps of different views. Figure 19 illustrates all the steps involved.

In practice, forming 3D cost volumes for each depth level and view requires large amounts of memory, which limits the resolution of the reconstructed shape. To mitigate this issue, R-MVSNet (Yao et al. 2019) replaces 3D CNNs with sequential 2D CNNs along the depth dimension using RNNs. This reduces the amount of required memory by 30% but at the cost of adding 25% to the runtime. MVSNet+ (CAS-MVSNet) (Gu et al. 2020) cascades

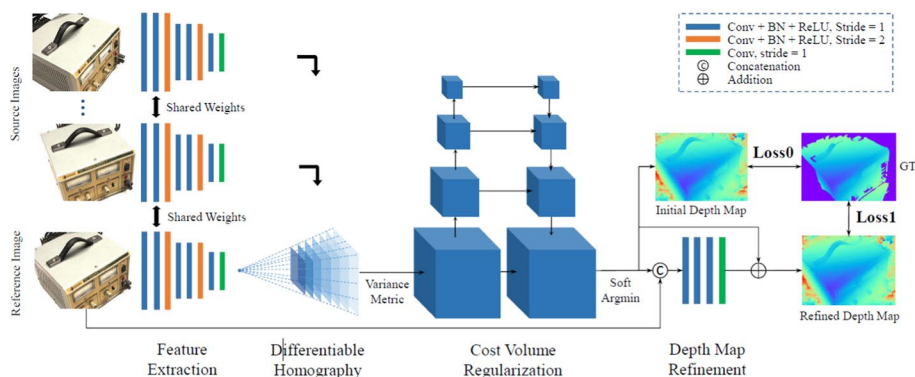


Fig. 19 Overview of MVSNet (Yao et al. 2018)

the formation process of cost volume in a coarse-to-fine-manner by adaptively choosing the number of depth hypotheses in each cascading stage. Compared to R-MVSNet, MVSNet+ improves runtime by about 65% and memory consumption by 25%.

Fast-MVS (Yu and Gao 2020) has a slightly lower runtime with respect to MVSNet+ and reduces memory consumption by 30%. The main idea behind Fast-MVS is to first infer a sparse cost volume and use it to construct a sparse depth map. In the second stage, a learned bilateral sampler helps propagate the sparse depth map into a dense one, considering reference image information. In the third stage, a differentiable Gauss-Newton layer refines the depth map further. To decrease the runtime and memory consumption of the previous methods, PatchmatchNet (Wang et al. 2021b) proposes a learned cascade formulation of the Patchmatch algorithm (Barnes et al. 2009) instead of using a plane sweep-based approach.

Tables 1 and 2 provide a performance comparison of the reviewed MVS methods and the most common public datasets, respectively.

9 How GANs can help?

The adversarial training strategy in GANs contributes to high-quality outputs. The generator and discriminator networks are jointly trained with an adversarial loss in a way that each tries to fool the other. The generator tries to produce more real outputs with higher

Table 1 Performance comparison of the reviewed MVS methods

Method	Acc. (mm)	Comp	Overall (mm)	Runtime (s)	Memory (GB)
SurfaceNet	0.450	1.040	0.745	N.M	N.M
MVSNet	0.396	0.527	0.462	1.25	11
R-MVSNet	0.383	0.452	0.417	1.70	7.5
MVSNET+	0.325	0.385	0.355	0.62	5.5
Fast MVSNet	0.336	0.403	0.370	0.55	4.1
PatchMatchNet	0.427	0.277	0.352	0.2	2

The results are obtained from Wang et al. (2021b). Acc and Comp are shorthands for accuracy and completeness, respectively

Table 2 The commonly used public datasets by MVS methods

Dataset	Image Res	Setting	GT type
DTU (Jensen et al. 2014)	2 Mpx	Realistic	Point Cloud + Cam. parameters
Tanks and Temples (Knapitsch et al. 2017)	8 Mpx	Realistic	Point Cloud
ETH3D (Schops et al. 2017)	0.4/24 Mpx	Realistic	Point Cloud + Cam. parameters
BlendedMVS (Yao et al. 2020)	0.4/3.1 Mpx	Synthetic	Depth + Cam. parameters

quality, while the discriminator tries to enhance its discrimination ability between fake and real samples. The adversarial criterion is known to perform better at capturing the structural difference of 3D objects than other criteria such as IoU or Chamfer distance, which can be misleading in some cases, leading to over-fitting. Another advantage of using GANs for 3D reconstruction is that one can take advantage of the representational capabilities of a trained discriminator and use it for the purpose of 3D shape classification, as done by 3D-GAN (Wu et al. 2016).

3D-GAN encodes an input image into a latent representation using a Variational Auto Encoder (VAE). The latent vector is then fed into the proposed 3D-GAN model. The model has a generator (as the decoder of the latent vector) with five 3D convolutional layers, which receives a latent vector and outputs the volumetric 3D shape in 64^3 resolution. The discriminator is a mirrored version of the generator, thus having five convolutional layers to infer the predicted shape as being real or fake. The loss function is comprised of a cross-entropy for the 3D-GAN model, a KL divergence loss for restricting the image encoder's output distribution, and an L2 reconstruction loss. However, the proposed method has the shortcoming of large memory consumption related to the use of a voxel grid representation, limiting the output resolution.

In Yu (2019), the authors propose a semi-supervised approach to 3D reconstruction using GANs. Given a set of 2D ground truth images with known camera poses along with an initial rough 3D reconstructed model, the generator refines the shape (represented in meshes) in each training step by applying a set of residual graph convolutional blocks. The discriminator receives the 2D ground truth images as well as rendered images of the generated 3D model, known as “observed images”. It then applies a set of residual convolutional blocks to classify the observed images as real or fake. The obtained results on the Tanks and Templates dataset (Knapitsch et al. 2017) show improved performance compared to MVS methods such as COLMAP (Schonberger and Frahm 2016). Since the output of this method is represented in meshes, it is ready to be used in many applications. However, this method needs a rough 3D model as input, which is obtained with other algorithms like spatial mapping (Pillai et al. 2016). Therefore, the quality of this method depends on the quality of the initial reconstruction.

Research (Pan et al. 2020) proves that pre-trained GANs on 2D images contain rich 3D knowledge about the object categories on which these models were trained. The researchers propose an unsupervised approach to single-image 3D reconstruction. As a starting point, the object's shape is initialized with an ellipsoid. A differentiable renderer is then used to render some pseudo-samples with varying lighting conditions and viewpoints. Since the shape is not yet refined, the rendered pseudo-samples are undesirable. To optimize the shape, one must have ground truth for each pseudo-sample. In that case, GAN inversion can help by predicting the latent offset for each pseudo-sample corresponding to the view changes and lighting conditions that were applied to the original image. The

inverted code is then used to recover the original image as ground truth. Therefore, the GAN inversion procedure is performed on the pseudo samples to get the projected samples. These samples are used as ground truth for the rendering process to optimize the 3D shape. These steps are performed iteratively to produce the final, refined shape. Figure 20 depicts the algorithm's framework. For more information about GAN inversion, please refer to Xia et al. (2022).

Summary: In conclusion, the adversarial training strategy in GANs contributes to a more detailed shape reconstruction, especially when the discriminator judges between estimated and ground truth 3D shapes. On the other hand, training such networks has a large memory footprint, limiting the reconstruction resolution as in 3D-GAN. 3D ground truth data is not always available for training learning-based methods. Moreover, obtaining 3D ground truth shapes is also time-consuming. In such cases, GANs can help to design a semi-supervised or even an unsupervised algorithm as in Yu (2019); Pan et al. (2020).

10 Beyond 3D reconstruction

Some studies have used 3D reconstruction as a downstream task to accomplish other goals. These methods mainly belong to the fields of object pose estimation, novel view prediction, 3D semantic segmentation, color reconstruction, and estimating lighting conditions for rendering synthetic objects in a scene. In the following, some of these methods are explained.

10.1 Novel-view synthesis

SynSin is a novel view prediction algorithm presented in /citep{wiles2020synsin}. The proposed model predicts a depth map (D) from an RGB image and extracts image features (F). With these two elements as inputs and a transformation (T), a novel differentiable renderer generates a point cloud of features representing the scene's 3D structure. As well as allowing gradient propagation, the proposed differentiable renderer generates features instead of RGB colors (unlike traditional techniques). A refiner network (generator) refines and inpaints the incomplete regions of the target view conditioned on the input image. It should be noted that cost functions are only applied to the output image and not to the point cloud. A discriminator network is also trained jointly with the generator adversarially to improve prediction quality. Figure 21 visualizes these steps.



Fig. 20 The reconstruction framework, as proposed and illustrated by Pan et al. (2020). An initial ellipsoid is refined iteratively by rendering some “pseudo samples” with lighting and viewpoint variations and applying GAN inversion to produce ground truth data for the rendering process. This process is performed iteratively to refine the 3D shape

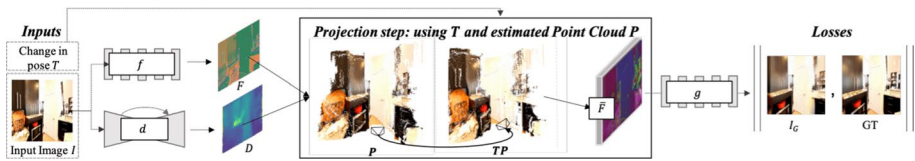


Fig. 21 Overview of SynSin (Wiles et al. 2020). Firstly, it estimates the 3D structure of the scene represented by a point cloud. The novel view of the scene is obtained by applying the desired transformation to the point cloud and rendering the result

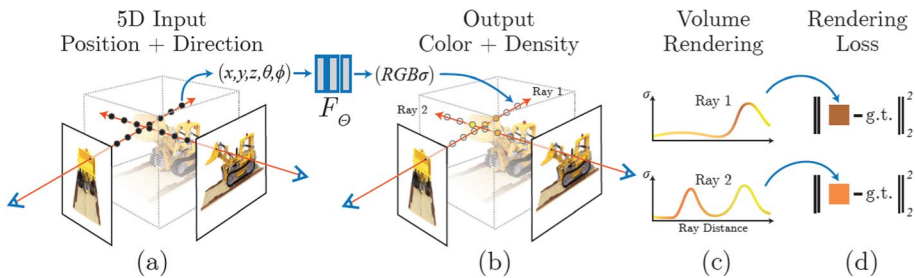


Fig. 22 An overview of neural radiance field scene representation and differentiable rendering procedure

10.1.1 Neural radiance field (NeRF)

With the introduction of NeRF (Mildenhall et al. 2020), a completely new direction for novel-view synthesis has been established. Given a sparse set of images capturing an object of interest from different viewpoints, NeRF learns an implicit function to represent complex 3D shapes such that the novel poses of the object can be synthesized. The implicit function is learned by an MLP. As illustrated in Fig. 22, the input to the function is a 5D vector consisting of 3D point coordinates (x, y, z) and a viewing direction (θ, ϕ) . The output is a 4D vector comprised of an emitted color (r, g, b) and a volume density σ . However, as the authors suggest, various aspects need improvement. The training and inference times of the original NeRF are high; therefore, instead of representing the entire scene as a single implicit field, Neural Sparse Voxel Fields (NSVF) (Liu et al. 2020) organize the scene as a sparse voxel octree and bound a set of implicit functions to its voxels. This speeds up rendering by a factor of ten. While NeRF only operates on static scenes, there are some other methods proposed to handle dynamic ones. Among them are Nerfies (Park et al. 2021), Space-Time Neural Radiance Fields (Xian et al. 2021), and NeR-Flow (Du et al. 2021). For instance, Given a video, “Nerfies” learns a deformation field for each video frame in addition to a color-density using a second MLP.

10.2 Color reconstruction

Some studies, including (L Navaneet et al. 2019) and (Liu et al. 2019), reconstruct not only the shapes of objects but also the colors of different parts of the inferred 3D shape. For example, L Navaneet et al. (2019) solve the task using regression. A differentiable module is used to project the predicted color for each point in the inferred point cloud onto a 2D image. Then the L2 distance between the predicted color and the ground truth is calculated. On the other hand, Liu et al. (2019) solve the problem via classification. Where, depending

on the representative colors present in the 2D input image, a set of colors is chosen by a network as a palette. Another network samples points from the 2D image for coloring. Then, according to the learned palette, the final color for each sampled point is recovered.

10.3 3D semantic segmentation

In some applications, such as robotics, the robot needs to gain a rich understanding of the scene. 3D semantic segmentation provides the robotic agent with valuable data to analyze and interact with the environment. Some works, including (Murez et al. 2020), (Zhao et al. 2017), and (L Navaneet et al. 2019), in addition to estimating the 3D shape of the scene, segment different parts and surfaces of the scene based on their semantics. This is believed to enhance the model's ability to infer 3D shapes more accurately as the supervisory signal now distinguishes between different surfaces and shapes.

10.4 3D scene captioning

Introduced by Chen et al. (2021), this task refers to joint 3D localization and generation of natural language descriptions for each of the present objects in a scene. For example, Chen et al. (2021) use an RGB-D scan as input and describes the localized objects in natural language. The relations between the localized objects are encoded in a graph. An MLP is responsible for extracting enhanced relational features from the graph. With the help of the attention mechanism, a captioning module generates words to form a description. An example of a captioned scene by this method is provided in Fig. 23.

11 Loss functions and evaluation metrics

This section lists and explains common loss functions for training deep learning-based 3D reconstruction algorithms and standard evaluation metrics. The loss functions are categorized based on the model's output representation. These categories include volumetric, point, and mesh-based losses for 3D supervision as well as loss functions used for 2D supervision.



Fig. 23 An example of 3D localization and captioning of a scene based on the visual features of the present objects as well as the physical relations between them, generated by Chen et al. (2021)

11.1 Volumetric based losses

L_2 distance: It is defined as the Euclidean norm calculated between ground truth and predicted volumes.

11.2 Negative intersection over union

This function is defined as the ratio of the overlapping area of both predicted and target volumes to the area of union (Eq. 4). This loss function is suitable for both binary occupancy and TSDF representations.

$$\mathcal{L}_{IoU} = -\frac{V_{pred} \cap V_{target}}{V_{pred} \cup V_{target}} \quad (4)$$

11.3 Cross-entropy

This function can be used in both binary and probabilistic occupancy outputs and is defined as:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N (p_i \log \hat{p}_i + (1 - p_i) \log (1 - \hat{p}_i)) \quad (5)$$

Where N is the number of voxels and $(p_i, \hat{p}_i)$ are predicted and ground truth occupancy probability of corresponding voxels.

11.4 Point based losses

11.4.1 Chamfer distance

Given predicted and target point sets, as P and Q , Chamfer Distance (CD) finds the distance between each point to its nearest neighbor in the other point set and sums up the results. In mathematical notation, the CD is given by:

$$\sum_{p \in P} \min_{q \in Q} \|p - q\|_2^2 + \sum_{q \in Q} \min_{p \in P} \|q - p\|_2^2 \quad (6)$$

11.4.2 Earth mover's distance (EMD)

The EMD solves an optimization problem to assign each target point q in set Q to a predicted point p in set P . This can be put in mathematical form as follows:

$$\min_{\phi: Q \rightarrow P} \sum_{q \in Q} \|q - \phi(q)\|_2 \quad (7)$$

where $\phi(q)$ is a set of one-to-one correspondences.

11.5 Mesh based losses

Models that generate an object's shape in a mesh representation have similar loss functions to point-based methods. These methods calculate both CD and EMD between ground truth and predicted mesh vertices. In addition to these losses, a normal loss is also applied. Nevertheless, these loss functions are not enough for the model to produce satisfying results. More precisely, the model easily gets stuck in local minima during training. Some studies consider regularization terms such as Laplacian and edge length to mitigate the issue. The regularization terms are explained in more detail below.

11.5.1 Normal loss

This loss function requires the edges between a vertex and its neighbors to be perpendicular to the corresponding surface normal from ground truth (Wang et al. 2018a). It can be described in mathematical notation as follows:

$$\sum_p \sum_{q=\text{argmin}_q(\|p-k\|_2^2)} \left\| \langle p-q, n_q \rangle \right\|_2^2, t \ k \in \mathcal{N}(p) \quad (8)$$

Where p is a predicted mesh vertex, q is the closest vertex to p that is found when calculating the chamfer loss, and k is the p 's neighboring vertex that belongs to the set of p 's neighboring vertices ($\mathcal{N}(p)$), $\langle \cdot, \cdot \rangle$ is the inner product of two vectors, and n_q is the observed surface normal from the ground truth.

11.5.2 Regularization terms

Two commonly used regularization terms for maintaining reconstruction quality are explained in the following:

11.5.2.1 Laplacian regularization Some studies including (Wang et al. 2018a) deform an initial mesh in multiple stages to infer a refined 3D mesh model. In order to prevent the 3D shape from becoming over-deformed, laplacian regularization is used after each deformation stage. It is defined as $\sum_p \left\| \delta'_p - \delta_p \right\|_2^2$ Where δ_p and δ'_p are Laplacian coordinates of a vertex before and after a deformation stage.

11.5.2.2 Edge-length regularization This regularization term prevents the model from producing irregular long edges. It can be defined as: $\sum_p \sum_{k \in \mathcal{N}(p)} \|p - k\|_2^2$.

11.6 Losses for 2D supervision

There are loss functions that allow the training procedure to be supervised in 2D rather than 3D. This leads to lower memory and resource consumption, so the output

resolution can be increased. The 2D image is rendered at some desired viewpoint using a projection operator and compared to the ground truth image to measure the loss. The image may be a silhouette, a depth map, or a combination of both. However, for the training procedure to be end-to-end, the projection operator must be differentiable.

11.6.1 Silhouette loss

This loss function measures the distance between an object's ground truth 2D silhouette (G) and the silhouette derived from the projection of the inferred 3D shape at the desired camera pose and intrinsic characteristics (P). The distance function d can be L_2 loss, Jaccard index, or the binary cross-entropy. It is defined as below:

$$\mathcal{L}_{Silhouette} = \left(\frac{1}{n}\right) \sum_{i=1}^n d(P^{(i)}, G^{(i)}) \quad (9)$$

where n is the number of silhouettes for each 3D model.

11.6.2 Render and compare

This loss function was first introduced by Kundu et al. (2018). In addition to providing 2D supervision, it is resource-efficient. It is based on calculating the IoU between predicted (P_s) and ground truth (G_s) silhouettes as well as the L_2 distance between rendered (P_d) and ground truth (G_d) depth maps. In mathematical notation, it can be described as follows:

$$\mathcal{L}_{RaC} = 1 - J(P_s, G_s; I_s) + MSE(P_d, G_d; I_d) \quad (10)$$

Here, J stands for the Jaccard index (segmentation IoU), and I stands for binary ignore masks. It is used to mask out pixels that do not contribute to the computation of the loss function.

11.7 Evaluation metrics

Evaluation metrics provide us with a quantitative assessment of how well the reconstruction algorithms perform individually and in relation to each other. The most commonly used evaluation metrics for the task of 3D reconstruction are IoU for voxel representation and both CD and EMD for point cloud and mesh representations. The F1-Score is also known as a robust metric for evaluating reconstruction quality. It is defined as the harmonic mean of precision and recall and can be expressed mathematically as:

$$F1\text{-Score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

Where precision and recall are calculated based on the percentage of sampled points in a prediction or ground truth that can find the nearest neighbor from the other within a certain threshold τ .

12 Datasets

Table 3 summarizes the characteristics of commonly used datasets in the field of 3D reconstruction. These datasets were published from early 2014 to 2020. ShapeNet (Chang et al. 2015) is the most used dataset in this field and Pix3D (Sun et al. 2018) has recently gained attention.

13 Comparison and discussion

As mentioned at the beginning of this study, traditional algorithms for 3D reconstruction require multiple posed images captured by well-calibrated cameras to infer the shape of the object(s). In addition, to achieve acceptable accuracy, the angular difference between the captured images must be small. There are no such restrictions for deep learning algorithms. These algorithms are even capable of reconstructing from a single image. However, the accuracy may be lower compared to the case when multiple posed images of the object(s) are present.

The type of output representation has a significant role in 3D reconstruction algorithms. It greatly impacts the network structure, performance of the algorithm, and application coverage. Because representation depends on the application, it is difficult to determine which is most appropriate.

The occupancy grids easily fit into the deep learning framework as one can generate them by performing a set of 3D convolutional operations on the features of the input 2D image. However, due to the use of 3D convolutions, it requires high resources. Consequently, reconstruction accuracy and resolution may be reduced. It is also important to note that this representation is not readily usable and needs some post-processing steps to prepare for real-life applications. For example, one can easily convert this representation to a mesh-based representation by predicting TSDF values at each voxel. Nevertheless, it is more challenging to predict continuous TSDF values than binary ones.

Another representation of 3D shapes is the point cloud which requires fewer resources compared to the voxel-based representation. Therefore, the reconstruction accuracy and quality will be better at a constant resource level. The point cloud models are more flexible and can be easily deformed. On the other hand, the generated shapes in the point cloud cannot be directly used in many applications and need to be transformed into a surface-based representation. In such cases, one may prefer to generate surface-based representations, such as meshes directly.

Mesh-based representations take up less memory than occupancy grids since they only model the surface of objects. However, the main problem with these representations is that they do not easily fit into deep learning frameworks. Another issue is that some generated mesh models are not manifold. This prevents precise reconstruction and requires special regularizers or additional processing steps to be considered.

Implicit Neural Representation is a powerful paradigm that parameterizes a signal as a continuous, differentiable function by neural networks. This newly emerged representation can reconstruct objects in high resolution with significantly reduced memory usage without the need for any post-processing steps and without the presence of issues like self-intersecting meshes or non-closed meshes. According to qualitative and

Table 3 Commonly used datasets for 3D reconstruction

Name	# samples	Resolution	# Obj	Type	Background	# Cls.	Camera Param.	GT type
KITTI12 (Geiger et al. 2012)	42K	1240×376	> 1	Real	Cluttered	2	All	Point-cloud
NYUv2 (Silberman et al. 2012)	1450	640×480	> 1	Real, Indoors	Cluttered	894	Intrinsic	Depth
Pascal3D+ (Xiang et al. 2014)	30.9K	Variable	> 1	Real	Cluttered	12	All	3D model
ShapeNetCore (Chang et al. 2015)	51K	Variable	1	Synthetic	Monotonic	55	Intrinsic	3D model
ModelNet40 (Wu et al. 2015)	12.3K	Variable	1	Synthetic	Monotonic	40	Intrinsic	3D model + P.C.
ObjectNet3D (Xiang et al. 2016)	90.1K	Variable	> 1	Real	Cluttered	100	All	3D model
ScanNet (Dai et al. 2017)	2.5 M	640×480	> 1	Real, Indoors	Cluttered	296	All	Mesh
SUNCG (Song et al. 2017)	130K	Variable	> 1	Synthetic	Cluttered	84	Intrinsic	Depth, Voxel
PIX3D (Sun et al. 2018)	9531	Variable	1	Real, Indoors	Cluttered	9	All	3D model
ABC (Koch et al. 2019)	1.0 M	1150×1033	1	Synthetic	Monotonic	N.M.	Intrinsic	3D CAD
Things3D (Xie et al. 2020)	1.68 M	256×256	1	Real	Cluttered	21,000	All	3D CAD

All both intrinsic and extrinsic camera characteristics are provided, *P.C.* Point Cloud, *N.M.* not mentioned

quantitative evaluations, the methods that use implicit representations have achieved state-of-the-art performance. However, they mostly have high inference times.

Other representations are not as common as the previous ones. For example, in Shin et al. (2019), researchers have used a multi-layered depth structure to represent a 3D model. This representation enables high-resolution reconstruction while consuming fewer resources than occupancy grids.

As previously noted, training 3D reconstruction models with 2D supervision has some advantages over 3D supervision, such as resource efficiency and simplicity. However, a differentiable rendering pipeline is needed. Since the 3D to 2D projection causes ambiguities, models trained with 2D supervision often fall a little behind the 3D supervision methods in terms of performance.

Shin et al. (2018) have conducted a study on the effect of output representation on the generalization of 3D reconstruction methods. The study shows that models that use multi-layer depth representation or 2.5D sketches (depth + segmentation), generalize better to new objects from the same categories or unseen objects of novel categories, compared to voxel representation. The mentioned study also prefers training models that predict 3D structures of objects in viewer-centered coordinates. It is worth noting that viewer-centered coordinates represent an object's shape in its original pose, while object-centered coordinates represent objects in their canonical pose. Therefore, in object-centered prediction, different views of objects in the input image are mapped to a single prediction, whereas in viewer-centered prediction, the task is more complicated for the model as it must predict the object's pose as well. This is a reason for the higher generalization of viewer-centered prediction (Shin et al. 2018). Tatarchenko et al. (2019) also suggest using viewer-centered coordinates and prove that neural networks trained for single image 3D reconstruction do not actually perform reconstruction but image classification. This fact is proved by evaluating a set of baseline recognition algorithms, including clustering, retrieval, and Oracle Nearest Neighbor. They show that these simple baselines yield better qualitative and quantitative results than state-of-the-art methods. They performed the Kolmogorov–Smirnov (Massey Jr 1951) test on the IoU histograms for all classes and all pairs of methods with the null hypothesis that two distributions exhibit no statistically significant difference, and found that for the deep learning-based methods and recognition baselines, the null hypothesis cannot be rejected, meaning that the former methods are statistically indistinguishable from baseline methods.

The pros and cons of the reviewed works are provided in Table 4. Table 5 summarizes the reviewed works in this article. Figures 24 and 25 compare the performance of the reviewed methods in terms of IoU and CD respectively, on ShapeNet. In Fig. 26 we compare the scaling capability of multi-view 3D reconstruction methods that were reviewed. Although the recently proposed methods for multi-view 3D reconstruction by Wang et al. (2021a) demonstrate a slightly better scaling capability than previous methods, we believe there is still more room for improvement for future works when providing more than 8 views of an object, especially when the object's structure is complex.

14 Conclusion and future research directions

In recent years, deep learning-based methods have achieved state-of-the-art results in the task of 3D reconstruction. Simultaneously, extensive and continuous research is being done to improve their performance. These methods have many advantages over

Table 4 Pros and cons of the reviewed 3D reconstruction algorithms

Name	Pros	Cons
3D-R2N2 (Choy et al. 2016)	Infers the shape from single and multiple views	Needs Object bounding box as input, Slow, limited to 32^3
OGN (Tatarchenko et al. 2017)	Supports high resolutions, low memory consumption compared to other voxel based methods	Binary output
Pix2Vox (Xie et al. 2019)	Easy to implement	Low output Res., Only single-view, Binary
Pix2Vox++ (Xie et al. 2020)	Supports multi-view reconstruction	Binary
VoiT (Wang et al. 2021a)	Hugely lower parameters, slightly better scaling capability	Binary, limited resolution
PSGN (Fan et al. 2017)	Easy implementation	Low output resolution
Lin et al. (Lin et al. 2018)	Dense output, 2D supervision, efficient training	Some 3D models lack details
3D-LMNET (Mandikal et al. 2018)	Easy to implement, S.t.A. results	2 training stages, auto-encoder knowledge is limited to train data
SGPCR (Zou and Hoiem 2020)	Handles object occlusion, better EMD than 3D-LMNET	Presence of a post-refinement step
Pixel2Mesh (Wang et al. 2018a)	Multiple losses for robust prediction, surface based output	Only single view, non-manifold meshes still exist
Pixel2Mesh++ (Wen et al. 2019)	Supports multi-view, lower parameters than Pix2Mesh	Existence of non-manifold meshes
TMN (Pan et al. 2019)	Prunes errors and refines boundaries	Presence of non-closed meshes, only single-view input
NMF (Gupta and Chandraker 2020)	Generates meshes with manifoldness	Over-smoothed meshes
ONET (Mescheder et al. 2019)	Arbitrary output resolution, efficient training	Only single-view input, high inference time
IM-NET (Chen and Zhang 2019)	Arbitrary output resolution, efficient training	Only single-view input, high inference time
DeepSDF (Park et al. 2019)	Arbitrary output resolution, efficient training	Only single-view input
3D-RCNN (Kundu et al. 2018)	Support for multiple objects	Needs B.B input, uses prior knowledge from limited data, high inference time
Mesh-RCNN (Gkioxari et al. 2019)	Surface based output, Multiple object support	Reconstruction of all objects depends on object detection accuracy
EFT-Net (Shin et al. 2019)	Higher generalization, surface based	Complex implementation
Atlas (Murez et al. 2020)	Real-time and efficient, entire scene Rec., semantic segmentation	Uniform accumulation of features along a ray, individual Obj.s lack details
CoReNet (Popov et al. 2020)	Handles Multiple objects, high Res., constant memory footprint	High inference time
Points2Objects (Engelmann et al. 2021)	Handles Multiple objects, real-time	Trained for shape retrieval not reconstruction

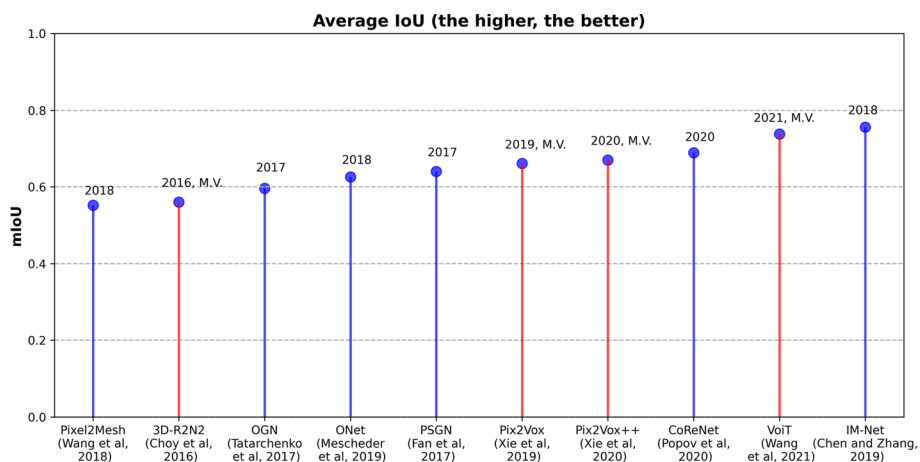


Fig. 24 mIoU of different 3D reconstruction algorithms on ShapeNet dataset (Chang et al. 2015). For IM-Net (Chen and Zhang 2019) mIoU is only reported for five classes. M.V., O.C., and V.C denote Multi-view input, object-centered, and viewer-centered prediction, respectively

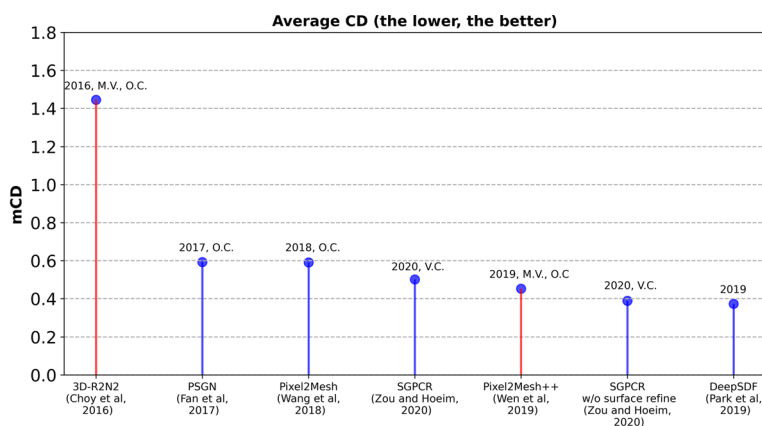


Fig. 25 Mean Chamfer distance of different 3D reconstruction algorithms on ShapeNet dataset (Chang et al. 2015). For DeepSDF mCD is only reported for five classes. M.V., O.C., and V.C denote Multi-View input, Object-Centered, and Viewer-Centered prediction, respectively

traditional ones, including simplicity, which eliminates the need for hand-crafted stages to achieve impressive results.

Different output representations have been used to embody 3D models in deep learning algorithms, including binary and probabilistic occupancy grids, point clouds, surface-based, and, more recently, implicit representations. As we have mentioned, the choice of output representation is a factor that profoundly affects network structure and reconstruction quality. Therefore, each representation has its own advantages and disadvantages and must be carefully selected depending on the application.

One limitation of the existing algorithms is that the trained models cannot generalize well to unseen object categories. There are limited solutions offered for this problem,

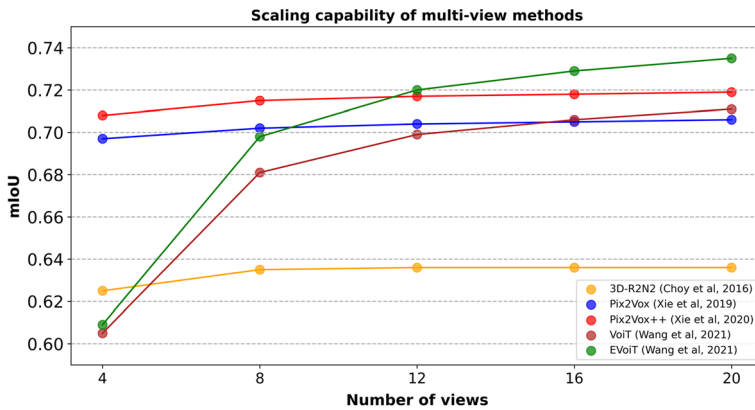


Fig. 26 The scaling capability of multi-view 3D reconstruction methods. The results are obtained from Wang et al. (2021a)

including predicting in the viewer-centered coordinates as well as the use of 2.5D output representation. However, these solutions have not entirely solved the problem. So we think future work should address this issue.

Few-shot learning seems to help in situations with limited available training data while one needs to obtain a relatively high generalization. For example, research (Wallace and Hariharan 2019) is dedicated to solving this problem.

Another limitation is present in the ill-posed task of single-view 3D reconstruction. As we mentioned in earlier sections, based on the findings of Tatarchenko et al. (2019), the deep learning methods for single-view reconstruction are biased towards recognition and retrieval instead of reconstruction. This problem could have three main reasons. The first reason refers to the prediction of the 3D shape in the canonical coordinates that can be mitigated by predicting in viewer-centered coordinates as investigated by Shin et al. (2018). The second reason is related to the metric types used for evaluating and reasoning about the model's performance, which can be misleading in the case of choosing IoU or Chamfer distance. Here, F-Score can be a simple yet reliable metric to use. It intuitively indicates the percentage of correctly reconstructed surface area or points that lie within a certain distance to the ground truth. The third reason for this bias is related to the composition of the training dataset as stated in Tatarchenko et al. (2019). ShapeNet is the most used dataset in this field. Since every object in a class is aligned to a canonical reference frame, there are many shapes in a class that are similar. Therefore, if one chooses an arbitrary shape from the test set, there is always a very similar shape in the training set. As such, a model simply needs to retrieve a similar shape from the training set without needing to reconstruct the shape of the object. A possible future work is mitigating the dataset issue by gathering a completely new dataset with viewer-centered coordinates. Furthermore, based on the findings of Tatarchenko et al. (2019), the variance of the predictions is also high for single-view reconstruction methods, clustering, and retrieval baselines. This indicates an overfitting issue, which is dissatisfactory for deep learning models. The GANs can also be used to alleviate single image reconstruction ambiguity, as investigated by Pan et al. (2020) and, recently, by Cai et al. (2022).

In real-world tasks such as robotic navigation and manipulation and 3D scene understanding, we need to infer the 3D structure of multiple objects or even the entire scene.

Table 5 Summary of the reviewed 3D reconstruction algorithms

Name	Year	Input	Input count	Reconstruction capability	Output representation	Output resolution
MSDN (Eigen et al. 2014)	2014	RGB Image	Single	–	Depth map	240 × 320
MV3DS (Tatarchenko et al. 2016)	2014	RGB + θ	Single	Single object	RGB, Depth, Mesh	128 × 128 × 4
3D-R2N2 (Choy et al. 2016)	2016	RGB	Multiple	Single object	Voxel	32 ³
OGN (Tatarchenko et al. 2017)	2017	RGB	Single	Multiple objects	Voxel (OCTree)	up to 512 ³
Pix2Vox (Xie et al. 2019)	2019	RGB	Single / Multiple	Single object	Voxel	32 ³
Pix2Vox++ (Xie et al. 2020)	2020	RGB	Single / Multiple	Single object	Voxel	up to 128 ³
PSGN (Fan et al. 2017)	2017	RGB	Single	Single object	Point-cloud	N (default=1024)
Lin et al. (Lin et al. 2018)	2018	RGB	Single	Single object	Point-cloud	N
3D-LMNET (Mandikal et al. 2018)	2018	RGB	Single	Single object	Point-cloud	N (default=2048)
DensePCR (Mandikal and Radhakrishnan 2019)	2019	RGB	–	Single object	Point-cloud	16,384
SGPCR (Zou and Hoiem 2020)	2020	RGB	Single	Single object	Point-cloud	4096
Pixel2Mesh (Wang et al. 2018a)	2018	RGB	Single	Single object	Mesh	–
Pixel2Mesh++ (Wen et al. 2019)	2019	RGB	Multiple	Single object	Mesh	–
TMN (Pan et al. 2019)	2019	RGB	Single	Single object	Mesh	–
NMF (Gupta and Chandraker 2020)	2020	RGB	Single	Single object	Mesh	–
ONET (Mescheder et al. 2019)	2018	RGB + Query point	Single	Single object	Binary occupancy at query point	Arbitrary
IM-NET (Chen and Zhang 2019)	2018	RGB + Query point	Single	Single object	Binary occupancy at query point	Arbitrary
DeepSDF (Park et al. 2019)	2019	RGB + Query point	Single	Single object	SDF value at query point	Arbitrary
3D-RCNN (Kundu et al. 2018)	2018	RGB + Object B.B	Single	Multiple objects	3D CAD	–
Mesh-RCNN (Gkioxari et al. 2019)	2019	RGB	Single	Multiple objects	Object B.B + class + Mesh	–
EFT-Net (Shin et al. 2019)	2019	RGB	Single	Scene	Mesh + Voxel	–
Atlas (Murez et al. 2020)	2020	Posed RGB	Multiple	Scene	Mesh + Object semantic segmentation	–
CoReNet (Popov et al. 2020)	2020	RGB	Single	Multiple objects	Voxel+Mesh	Arbitrary

Table 5 (continued)

Name	Year	Input	Input count	Reconstruction capability	Output representation	Output resolution
Points2Objects (Engelmann et al. 2021)	2021	RGB	Single	Multiple objects	Based on the training data	Arbitrary
VoiT (Wang et al. 2021a)	2020	RGB	Multiple	Single object	Voxel	32 ³

However, most of the current research is dedicated to single-object reconstruction. Therefore, many aspects must be considered and improved, including the physical plausibility of the results, resource efficiency, and accuracy.

Recently, deep learning networks have been introduced that reconstruct objects using implicit neural representations. As these methods emerged, the need for high memory resources was eliminated. Nevertheless, optimizing these methods is usually time-consuming and challenging. We, therefore, expect to see more research on different optimization methods for these algorithms, like (Finn et al. 2017). In such methods, model-agnostic meta-learning is used to generate an initial guess for the network parameters so that the training process takes much less time and avoids over-fitting. Another notable direction in future research is eliminating expensive 3D supervision when training learning-based 3D reconstruction algorithms. The 3D supervision needs a considerable amount of 3D data, which is, to this date, hard to obtain. Moreover, 3D supervision is computationally expensive and consumes many resources. Recent studies consider applying 2D supervision by developing differentiable rendering modules that allow end-to-end training of 3D reconstruction methods with only 2D data to mitigate this issue. GAN inversion procedure accompanied by a differentiable renderer can contribute to developing 2D supervised 3D reconstruction methods. Based on the findings of Pan et al. (2020), the 3D knowledge within pre-trained 2D GANs can be utilized to lower the uncertainty of the single image 3D reconstruction task. Even in the 2D domain, the converging issue in training GANs has always been a challenge.

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